

# The conditions for a qudit advantage in uncorrected quantum algorithms under decoherence

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Whether qudits buy resilience against decoherence in full algorithms is open: prior noisy comparisons stop at gate-level criteria, arithmetic primitives, or a fixed dimension, and error-correcting codes have no instance to compress. We simulate Shor order finding, eigenstate phase estimation, and Grover search as bare, uncorrected circuits on qubit, qutrit, and ququint registers, under transmon-calibrated ladder and trapped-ion depolarizing channels, three entangling-gate cost models, and registers to Hilbert-space dimension  $5.3 \times 10^5$ . Two conditions organize all three. A qudit advantage needs an entangling gate whose cost grows at most linearly in  $d$ : two-level-decomposed gates forfeit it at  $d \geq 5$  on the ladder and at  $d = 7$  everywhere, and the decomposed qutrit’s residual advantage survives realistic accounting only under per-particle noise. And it survives only while the gate’s own infidelity growth with  $d$  stays below a critical inflation factor we compute: converted through each channel’s damage identity, the only native gate characterized across  $d = 2/3/5$  (99.6/98.7/93.7%) fails at  $d = 5$  in seven of eight readings, while its qutrit gate passes seven of eight; the transmon qutrit gate passes only when charged solely on the entangler, and loses once its 580-ns duration is charged as exposure. Further tightenings independently leave only  $d = 3$ . Accumulated channel damage fixes end-state fidelity on one exponential across algorithms and bases; what the algorithm adds is its decoder, for which we derive an exact finite-size mediant-interval acceptance law, verified outcome-for-outcome, base-independent at matched control dimension: the quantum state carries the whole cross-base difference. A number-theoretic confound, grid alignment, reverses naive cross-dimension comparisons. A shallow trapped-ion anchor fixes the channel’s scale; deeper anchors fail coherently, bounding the validated depth.

## I. INTRODUCTION

Quantum hardware is not binary by nature. Transmons, trapped ions, nitrogen-vacancy centers, molecular spins, and Rydberg-blockaded atom arrays all expose more than two usable levels, and processors that exploit them as qudits now exist or are proposed on several platforms [1–7]. The theoretical case for higher dimensions is old and, at the level of structure, strong: fault-tolerant stabilizer constructions close cleanly in prime dimension [8] and the discrete phase-space structure is cleanest in odd dimension [9]; the quantum arithmetic fractional Fourier transform of Ref. [10] exists for odd prime  $p$ , where the discrete rotation group  $SO_2[\mathbb{Z}_{p^n}]$  is cyclic (the in-place constant multipliers and quadratic-phase diagonals of the same construction carry over to any dimension, including  $p = 2$ , whenever the multiplication constant is a unit); magic-state distillation protocols exist in every prime dimension [11], with thresholds and efficiencies that keep improving with  $d$  (within congruence classes mod 3, for primes up to 17) [12]; and qudit memory thresholds rise under abstract noise models [13]. A base- $d$  register also compresses algorithms: at matched problem size it needs  $\log_2 d$  times fewer carriers and a proportionally shorter schedule.

Whether any of this survives *physical* decoherence—in which higher levels of a real device decay and dephase

faster than lower ones—is a question the literature has flagged repeatedly and not answered. Campbell noted that “in physical systems one may also see noise rise with  $d$ . Such features depend subtly on the details of the underlying physics” [12]; Marks *et al.* point to the “increased degrees of freedom that can be coupled to the environment” and call whether such noise can be kept sufficiently small “an interesting question for experimental implementations” [13]; Pavlidis and Floratos, citing qubit evidence that order finding tolerates truncation of small QFT rotations [14, 15], conjecture “a similar robustness” for qudits but leave its investigation open [16]; and the recent qudit review of Kiktenko *et al.* lists “the investigation of the impact of noise within the discussed schemes” as an open problem [17]. Comparisons *across* dimensions still treat noise indirectly: entangling-gate counts as a noise proxy [17, 18], analytic coherent-error bounds [19], or the qualitative argument that fewer carriers means less local noise [20]. Noisy qutrit circuit simulations exist [21, 22], but always at  $d = 3$  against a qubit baseline; the closest cross-dimension noisy study, a QFT-based adder under local noise across arbitrary  $d$  [23], compares a single arithmetic primitive rather than a full algorithm with its decoder. No study has compared several prime dimensions on full algorithms under channels calibrated to measured per-level coherences, or followed such a comparison to scale. A claim that qudit phase-estimation error decreases exponentially with  $d$ —attributed by the review of Wang *et al.* [20] to a conference contribution of Parasa and Perkowski [24], whose available slides state it for the algorithm’s intrinsic failure

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probability at fixed precision—concerns the noiseless algorithm; we are not aware of it having been tested against a noise model.

This paper answers the question quantitatively at the algorithm level. We simulate Shor order finding [25], eigenstate quantum phase estimation (QPE) [26], and Grover search [27] on qubit ( $d = 2$ ), qutrit ( $d = 3$ ), and ququint ( $d = 5$ ) registers, under two decoherence channels chosen to bracket real hardware—an anharmonic-ladder channel calibrated to published per-level transmon coherence measurements [2, 3, 28–30], and a per-particle depolarizing channel representative of trapped-ion qudits [1, 4]—under three entangling-gate cost models spanning native gates, the measured trapped-ion  $2(d-1)$  Mølmer-Sørensen cost [1], and  $O(d^2)$  two-level decomposition [16], with registers to Hilbert-space dimension  $5.3 \times 10^5$  (19.0 qubit-equivalents; the widest qubit register is 17 qubits) reached by quantum trajectories; a demo-size  $d = 7$  run (Sec. IV) extends the cost grid one prime higher.

Our central result is a pair of conditions rather than a verdict:

*Qudit gates compiled by two-level decomposition ( $O(d^2)$  layer cost on the entanglers) forfeit the advantage at  $d \geq 5$  on the calibrated ladder and at  $d = 7$  in every channel; at  $d = 3$  the decomposed circuit is still shorter than the qubit’s and the serial advantage survives both channels, though on the ladder it is erased by either measured single-qudit costs or concurrent scheduling, leaving only the per-particle qutrit, thinly (Sec. IV). A native entangling gate whose cost grows no faster than linearly in  $d$  delivers the advantage when the operating dephasing is weak or unstructured—but only while the gate’s own per-application infidelity growth with  $d$  stays below a critical inflation factor  $f^*$  (Sec. VIII). Measured hardware decides the second condition against the ququint in seven of eight readings and for the trapped-ion qutrit in seven of eight; the transmon qutrit’s verdict turns on whether a platform’s idle decoherence scales with its entangling-gate error, and on whether its entangler is charged in layers or in wall-clock time. Free-evolution ladder dephasing further splits the linear-cost cells by algorithm and dimension, and Zeeman-structured dephasing—higher levels dephasing tens of times faster—reverses the verdict at every cost model in its Markovian idealization, though not under laboratory quasi-static field noise (Secs. IV and VIII).*

The condition is a statement about bare, uncorrected circuits—the regime of every near-term demonstration. It does not transfer automatically to the error-correction layer, where a code has no problem instance to compress

and the dimension dependence can carry the opposite sign [31] (Sec. X).

The native-gate cost requirement is met by current trapped-ion qudit gates—the single-application light-shift entangler of Hrmo *et al.* [32] realizes it up to  $d = 5$ —and by native cross-Kerr transmon entanglers [3], and violated whenever qudit gates are compiled by two-level decomposition [16, 33]. The infidelity condition is what both measured gates currently fail at their  $d = 5$  frontiers: the ion gate’s error grows  $16\times$  from  $d = 2$  to  $d = 5$  ( $0.004 \rightarrow 0.063$ ), exceeding the critical inflation factor at  $d = 5$  in seven of eight readings, while its  $d = 3$  gate now clears the threshold under both charging scopes; the transmon cross-Kerr gate’s measured 97.3% clears the  $d = 3$  threshold only when the factor is charged solely on the entangler—0.9 points short of the global-scope target, and lost outright once its 580-ns duration is charged against a two-qubit comparator faster than  $\sim 250$  ns (Sec. VIII). Together the two conditions explain the split verdicts in prior work: qutrit advantages obtained with native third levels—as ancillas [21] or as encoding capacity [22]—versus 35–69% more non-Clifford gates for single-qutrit synthesis from a fault-tolerant gate set [33]. An independent analytic gate-level criterion [34], which we reproduce numerically to  $4 \times 10^{-4}$  and apply to our circuits under the identification of layer-count ratios with gate-efficiency ratios (an assumption scoped in Sec. IV), is consistent with our algorithm-level outcomes in four of six cost-model/dimension cases (all misses in the conservative direction), with break-even gate-efficiency ratios of 1.68 ( $d = 3$ ), 3.45 ( $d = 5$ ), and 5.70 ( $d = 7$ )—a constant factor of  $\sim 3$  below the naive  $d^2/\log_2 d$  estimate.

Three further results are methodological and, we believe, of independent use to the field.

(i) *Grid alignment is a confound in every cross-dimension comparison of order finding* (Sec. III). Phase estimation concentrates probability on the phases  $s/r$ ; when the order  $r$  divides the control dimension  $D = d^m$  these sit exactly on measurement grid points and the peaks are maximally noise-robust. Which base receives this gift is decided by number theory, not physics. The canonical demonstration instance  $N = 15$  admits *only* power-of-two orders, silently handing qubits perfect alignment; on unbiased instances the ordering of bases reverses. Alignment predicts the winner in five of the six biased runs we construct (three aligned instances  $\times$  two noise models); the exception is  $N = 15$  under depolarizing, where the aligned qubit trails the ququint by  $1.4\sigma$  ( $0.72 \pm 0.05$  against  $0.81 \pm 0.03$ ) while holding its lead in the ladder channel—a miss inside the noise of a 400-trajectory point, but a miss, and we count it as one. A within-modulus control—two orders hosted on identical registers—prices it at  $\approx 0.2$  in floor-corrected signal (a figure measured on prime-order alignment, where a corollary of the decoder lemma—noiseless aligned success is exactly  $\varphi(r)/r$ —shows the gift is largest; Sec. III).

(ii) *The mechanism is width-and-depth compression, and what survives it is the decoder* (Secs. VI–VII). Grover

search holds the oracle count fixed across bases, delivering roughly half of Shor’s exposure compression: the qudit advantage survives (confirming a prediction recorded in the source before the runs) at 0.33–0.50 of Shor’s. Counted in *damage units*—the per-carrier-layer entanglement infidelity  $1 - \text{tr} S/d^2$  of the actual channel—accumulated exposure places end-state fidelity for both algorithm families on a single exponential ( $R^2 = 0.97$ – $0.99$  in linear fidelity,  $0.74$ – $0.75$  rescored in log fidelity, amplitude  $\approx 1$ )—what first-order composition of incoherent channels requires, rather than a discovery. What exposure does *not* determine is the measured success probability, because the classical decoder intervenes: continued-fraction order recovery gains error tolerance with register size, and we observe it decoding correct orders from states with fidelity  $\sim 6 \times 10^{-4}$  to the ideal state. That tolerance has a closed form we state and verify—an exact finite- $D$  mediant-interval formula for the acceptance set of a decoder required to return  $r$  itself, built on a proved acceptance lemma and including the deeper convergents  $kr \leq N$  that window bounds do not count—reproducing the enumerated acceptance set outcome-for-outcome on 27 instance/size combinations, with the lemma checked on 42; its asymptotic density is a totient sum over the admissible denominators (Sec. VIIC, Appendix A). Because it turns on  $r$  and  $N$  and not on the base, at matched control dimension the whole cross-base difference in decoded success sits in the quantum state.

(iii) *Standing reviewer objections are pre-empted with data* (Sec. VIII):  $d$ -dependent readout error is structurally near-neutral (the growth of per-level misread rates with  $d$  cancels against the reduced carrier count), and refocused operation helps qudits *more* than qubits in all conditions tested, moving the break-even point of the central condition by roughly one cost model. The same section stress-tests the noise conventions themselves: against inflated per-layer qudit noise, against Zeeman-structured dephasing (where the condition genuinely fails unmitigated), and against composite dimensions  $d = 4$  and  $6$ .

Finally, the qubit branch of the predictions is checked on real hardware (Sec. IX): on a commercial trapped-ion processor the shallow compiled circuit lands inside its predicted band—a single-point consistency check of the depolarizing convention on the platform it models—while the deep-circuit and fixed-connectivity failures delineate, on hardware, the coherent-systematic and compilation-overhead regimes our channels deliberately exclude, bounding the convention’s validated depth range.

Prior work closest to ours is differentiated in Sec. X: the gate-level criterion of Janković *et al.* [34] (no algorithms, no scaling), the qutrit circuit studies of Gokhale *et al.* [21] and Gustafson [22] ( $d = 3$  only; no comparison across multiple prime dimensions, no per-level-calibrated channel), the noisy cross-dimension adder of Agrawal *et al.* [23] (one primitive, no decoder, no calibrated channel), and the code-level comparison of Keppens *et al.* [31],

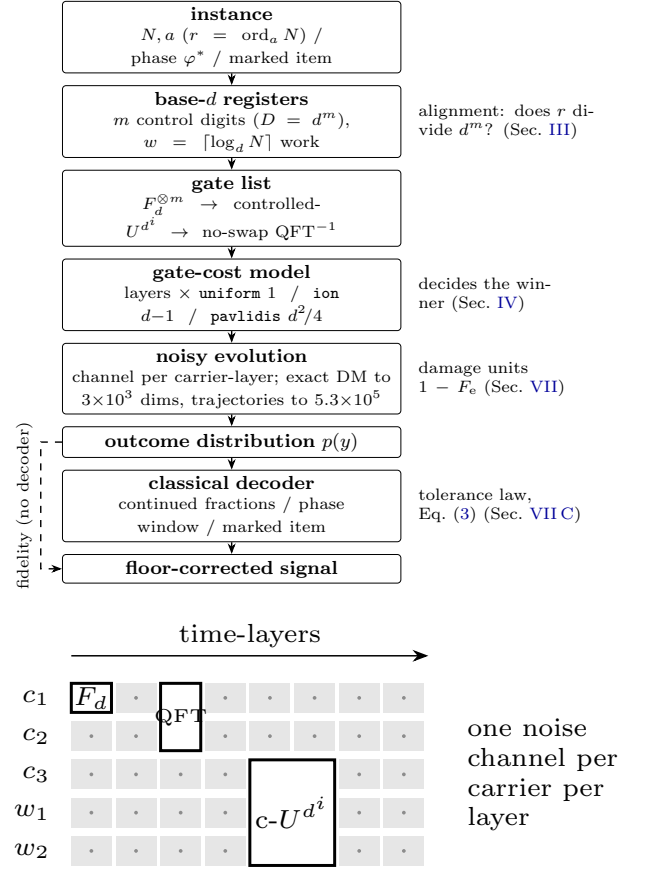


FIG. 1. Top: the simulation pipeline. Every algorithm, base, and channel flows through the same stages; the four annotations mark where the paper’s main objects enter—the alignment confound at register choice, the cost model that decides the winner, damage-weighted exposure in the noise stage, and the decoder whose tolerance obeys Eq. (5). The dashed path is end-state fidelity, which bypasses the decoder and obeys the one-curve damage law of Sec. VII. Bottom: the exposure convention. Rows are carriers (control  $c_i$ , work  $w_i$ ), columns are time-layers; every carrier passes through the noise channel every layer (dots), a gate spanning  $k$  carriers occupies its cost-model depth in layers (boxes), and idling carriers decohere while gates execute. Exposure = carriers  $\times$  layers; in damage units it is exposure  $\times (1 - F_e)$ .

whose opposite ordering we reconcile explicitly: a code has no problem instance to compress, so it pays the per-carrier cost of dimension without the width-and-depth rebate that is the entire source of the algorithmic advantage measured here.

## II. SETUP

Figure 1 summarizes the simulation pipeline and the exposure accounting that every result in this paper shares; the subsections below specify each stage.

## A. Algorithms and registers

We compare three algorithms chosen to span the failure modes of interference under decoherence. *Shor order finding*: given  $N$  and a coprime  $a$ , find the multiplicative order  $r$  of  $a \bmod N$ , using a control register of  $m$  base- $d$  digits (control dimension  $D = d^m$ ) and a work register of  $w = \lceil \log_d N \rceil$  digits; success is the probability that continued-fraction post-processing of the measured outcome recovers  $r$  exactly. Our benchmark instances are unbiased in the sense of Sec. III:  $N = 21$ ,  $a = 2$  ( $r = 6$ ) and  $N = 29$ ,  $a = 16$  ( $r = 7$ ). *Eigenstate QPE*: phase estimation on an eigenstate of a diagonal unitary with target phase equal to the golden-ratio conjugate  $\varphi^* \approx 0.6180$ , chosen to be far from every fraction with a small base-2, -3, or -5 denominator; success is the estimate landing within  $2^{-(b+1)}$  of the target ( $b = 5$  bits). *Grover search* over  $M = d^n$  items with  $\lfloor (\pi/4)\sqrt{M} \rfloor$  iterations; success is the probability of measuring the marked item, averaged over a sample of marked items excluding  $|0 \dots 0\rangle$  (the diffuser’s own reflection axis and the bottom of the damping ladder).

Because  $M = d^n$  never matches exactly across bases, all cross-base comparisons are made at matched problem size by interpolating each base’s results onto a common  $\log_2$ -size axis (linear interpolation of the signal in  $\log_2$  size); comparing raw demo-size points reverses orderings and is one of the two fairness traps we document (the other is alignment).

## B. Noise channels

One time-layer of noise is applied to *every* carrier for every layer of the serial schedule; a gate spanning  $k$  carriers occupies its decomposition depth in layers (below), so idling carriers decohere while gates execute.

*Calibrated ladder (transmon-like)*. A Lindblad channel with sequential relaxation  $\Gamma_k \propto k^{0.7}$  and pure dephasing following a max-level law  $\Gamma_\phi(j, k) \propto \max(j, k)^{1.1}$ , exponentiated exactly per layer. Both exponents are fits to published per-level coherence data spanning nine devices and  $d = 3$  to 12: the relaxation ratio  $\Gamma_2/\Gamma_1$  is measured at  $\approx 1.7$  [2, 3, 28–30, 35] against 2.0 for the textbook  $\Gamma_k \propto k$  ladder, and the dephasing ratios  $\Gamma_\phi^{01} : \Gamma_\phi^{12} : \Gamma_\phi^{02}$  are measured at  $1 : 2.0 : 2.3$ —incompatible with the textbook  $(\Delta\text{level})^2$  law (which predicts  $1 : 1 : 4$ ) because the charge dispersion of  $|2\rangle$  exceeds that of  $|1\rangle$  by an order of magnitude [2]. The max-level dephasing target is realized *exactly* by diagonal jump operators obtained through classical multidimensional scaling (Sec. XI); our channel reproduces the measured ratios as 1.62 and  $1 : 2.14 : 2.14$ . Rates are normalized so the  $0 \leftrightarrow 1$  subspace is bit-for-bit identical to the qubit channel: every difference between bases is purely a higher-level effect. A dephasing-scale knob interpolates to the high- $E_J/E_C$  regime demonstrated at  $d = 12$  [30], where echo coherence approaches the  $T_1$  limit; the same knob

models refocused operation (Sec. VIII).

*Per-particle depolarizing (ion-like)*.  $\rho \rightarrow (1-p)\rho + p\mathbb{I}/d$  per carrier-layer. The convention rests on measured trapped-ion structure: every qudit level is ground-state or metastable ( $\tau_1 \sim 1.1$  s, orders of magnitude beyond gate times), allowed-transition magnetic sensitivities span only a factor of  $\sim 5$  (pair dephasing rates go as the square, and idling pairs beyond the driven set widen the span—Sec. VIII prices the full anisotropy, a  $1\text{--}49\times$  rate spread at  $d = 5$ ), and the single-qudit *per-pulse* error is nearly flat in  $d$  ( $2.0 \times 10^{-4}$  at  $d = 3$ ,  $3.2 \times 10^{-4}$  at  $d = 5$ ; the per-*Clifford* error rises  $5\times$ ,  $2 \times 10^{-3}$  to  $1.0 \times 10^{-2}$ , because a  $d = 5$  Clifford takes  $\sim 3\times$  the pulses) [1], with qudit control demonstrated to 13 levels [4]. A Zeeman-structured dephasing variant that keeps the encoding’s full pair anisotropy is tested in Sec. VIII.

Measured per-gate infidelities place transmon two-qutrit gates at  $2.7 \times 10^{-2}$ – $4.8 \times 10^{-2}$  ( $\text{CZ}^\dagger/\text{CZ}$  process infidelity [3]) and ion two-qudit entangling gates at  $1.3 \times 10^{-2}$ – $6.3 \times 10^{-2}$  for  $d = 3\text{--}5$  (native light-shift [32]) and  $2.5 \times 10^{-2}$ – $6.2 \times 10^{-2}$  at  $d = 3$  (decomposed [1]); only single-qudit pulses reach the  $10^{-3}$ – $10^{-4}$  scale [1, 4]. These infidelities come attached to durations that differ by orders of magnitude across platforms—the transmon  $\text{CZ}^\dagger$  takes 580 ns, ion entanglers run on the sub-millisecond scale against proportionally longer coherence times—and Sec. VIII makes the duration side explicit by restating the threshold  $f^*$  as a wall-clock tolerance. In the per-carrier-layer convention above, a measured gate infidelity  $\epsilon$  over  $L$  layers implies  $s = \epsilon/[2L(1 - 1/d^2)]$ ; our sweeps (0.001–0.05 at demo size, 0.003–0.005 for scaling) bracket the implied operating points, which are marked on every figure, and Sec. VIII evaluates the grid at the per-base strengths the measured gates actually imply.

Four named noise regimes appear in the figures: the *idealized ladder* ( $\Gamma_k \propto k$ ,  $(\Delta\text{level})^2$  dephasing—the textbook model, used only as a bound), the *calibrated ladder* above, its *low-charge-dispersion* variant (the dephasing knob set to the high- $E_J/E_C$  regime of Ref. [30]), and *per-particle depolarizing*.

## C. Gate-cost models

Real hardware charges more per entangling gate as  $d$  grows, and this is the strongest objection to any qudit advantage. We charge three published cost structures as layer multipliers: **uniform** (native qudit entangler, one layer regardless of  $d$ —e.g. the cross-Kerr CZ of Ref. [3] or the single-application light-shift gate of Ref. [32]); **ion** ( $d - 1$  per entangling gate: the supplement of Ref. [1] decomposes a  $d$ -dimensional controlled-increment into  $2(d-1)$  pairwise fully entangling gates, which we normalize to 1 at  $d = 2$ , the  $d=2$  MS gate being itself the two-pulse unit—the same reference’s main text quotes  $2d$  MS gates for the Cinc and a *d-independent* two-gate cost for the controlled-exchange, so linear-in- $d$  is the representative middle of the measured constructions rather



than a unique count); and `pavlidis` ( $d^2/4$  on two-qudit gates, the harshest of the three: the *controlled* rotations of the QFT-arithmetic circuits of Ref. [16] decompose into  $4(d-1)^2$  elementary two-level gates, i.e.  $(d-1)^2$  per gate after the same  $d = 2$  normalization, and our  $d^2/4$  charge rounds that cost *down*—2.25 against 4 at  $d = 3$ , 6.25 against 16 at  $d = 5$ ; the same reference’s Table 1 carries the  $d^2$  factor in circuit *depth* as well ( $8d^2q$  for the QFT,  $4d^2q$  per adder), and the follow-up construction of Ref. [10] retains it ( $16p^2n$  for the in-place constant multiplier). The  $d^2$  is the two-level decomposition cost of a controlled rotation; a *single*-qudit rotation decomposes into  $O(d)$  two-level gates, not  $O(d^2)$ , so it stays at one layer here and its measured cost is priced separately in Sec. VIII). The three models swing the ququint Shor circuit from  $3.8\times$  *shorter* than the qubit’s (15 vs 57 layers on  $N = 21$ ) to  $1.1\times$  *longer* (62.2 vs 57)—and that swing, not the noise model, does most to decide the winner. These counts are *layers*, not wall-clock time; Sec. VIII charges measured gate durations and reports where that changes the verdict. Multi-qudit gates in Grover (oracle and reflection) are applied as exact unitaries but charged their  $n-1$ -layer decomposition depth; charging them one layer would hand the largest free ride to the base packing the most carriers into the gate, which is  $d = 2$ . All three models charge a *single*-qudit gate one layer at every  $d$ , which matches the per-pulse benchmarking of Ref. [1] but not its per-Clifford figures; since the single-qudit share of the schedule grows with  $d$  (21%/31%/40% at  $d = 2/3/5$ ), that omission favors qudits, and Sec. VIII charges it and reports the threshold at which each cell fails. Unless a figure states otherwise, plotted results use the `uniform` (native-entangler) cost model—the most qudit-favorable of the three; Table II and Secs. IV, VI, and VIII quantify how the winner changes under `ion` and `pavlidis`.

### D. Success metric

Raw success probabilities have base- and instance-dependent random floors: a uniformly random outcome passes continued fractions with probability 0.12–0.13 on the  $N = 21$  benchmark at demo size, 0.20–0.26 on  $N = 29$ , and up to 0.59 on the scorable small-order instances (at  $r = 2$  the floor reaches 0.72–0.80, meeting or exceeding the noiseless baseline). All comparisons therefore use the floor-corrected signal

$$\text{signal} = \frac{\text{success} - \text{floor}}{\text{success}_{\text{noiseless}} - \text{floor}}, \quad (1)$$

which is 1 for perfect interference, 0 for random guessing, and negative when noise actively biases outcomes away from the answer. Floors and noiseless baselines are recomputed for every base, size, and readout-error setting. Because small orders compress the metric’s dynamic range, we require a floor-to-baseline span above 0.15 for any instance used in a quantitative comparison; the one exception is the  $r = 3$  aligned instance of

Fig. 2 (spans 0.07–0.12), retained because no other instance represents its alignment class—its winner margins far exceed its metric compression, but we quote no signal magnitudes from it.

### III. GRID ALIGNMENT: A CONFOUND THAT DECIDES WINNERS

Phase estimation concentrates probability on the phases  $s/r$ . If  $r$  divides  $D = d^m$ , these sit exactly on grid points of the base- $d$  measurement basis and the interference peaks are perfectly sharp; otherwise they smear over neighboring outcomes, and smeared peaks are far more fragile under noise. Which base receives sharp peaks is decided by the arithmetic of the instance, not the physics of the hardware—and it is invisible in the existing literature because prior studies work at fixed  $d$ , where alignment is a constant.

The canonical instance  $N = 15$  is maximally pathological: its multiplicative group is  $\mathbb{Z}_2 \times \mathbb{Z}_4$ , so *every* order is a power of two and the qubit register always lands exactly on grid while  $d = 3, 5$  never do. Any cross-dimension comparison run on  $N = 15$ —including the first version of this study—hands base 2 a structural gift unrelated to decoherence.

Figure 2 shows one instance per alignment class. The aligned base wins in **five of six** biased runs, with margins exceeding anything decoherence produces between bases (at  $r = 5$  the aligned ququint scores 1.00 against the qubit’s 0.25–0.43); the exception is  $N = 15$  under depolarizing noise, where the aligned qubit trails the ququint by  $1.4\sigma$  ( $0.72 \pm 0.05$  against  $0.81 \pm 0.03$ ) while holding its lead in the ladder channel. On the two unbiased instances ( $r = 6, r = 7$ ) all four runs put both qudits above the qubit at  $\geq 5\sigma$ ; the  $d = 5 > d = 3$  leg is decisive under depolarizing ( $3.9\sigma$  and  $4.1\sigma$ ) but a statistical tie on the ladder ( $1.1\sigma$  at  $N = 21$ ,  $0.3\sigma$  at  $N = 29$ ). Misalignment is moreover graded, not binary: averaging the distance of the phases  $s/r$  to the nearest grid point, the  $r = 6$  instance is mildly tilted *toward* the qubit (0.267 vs 0.300) and the qubit still loses by 0.21–0.48, while  $r = 7$  ( $N = 29$ ) is exactly neutral across all three bases—the cleanest instance we know of, and the one we recommend for future benchmarks.

*Within-modulus control.* Varying  $N$  also varies register widths, so we isolate alignment with two moduli that each host an aligned and an unaligned order on *identical* registers:  $N = 33$  and  $N = 55$  carry both  $r = 5$  (exact in base 5) and  $r = 10$  (exact in no base) with the same  $(m, w)$ , gate counts, and noise exposure per base. Removing alignment costs the ququint 0.15–0.22 signal and narrows its lead over the qubit by 0.09–0.21, consistently across both moduli and both noise models—while the ququint’s *residual* lead on the unaligned orders remains 0.37–0.59, roughly twice the alignment term. Both effects are real; alignment is worth on the order of 0.15–0.2 signal to whoever receives it, and the remaining qudit

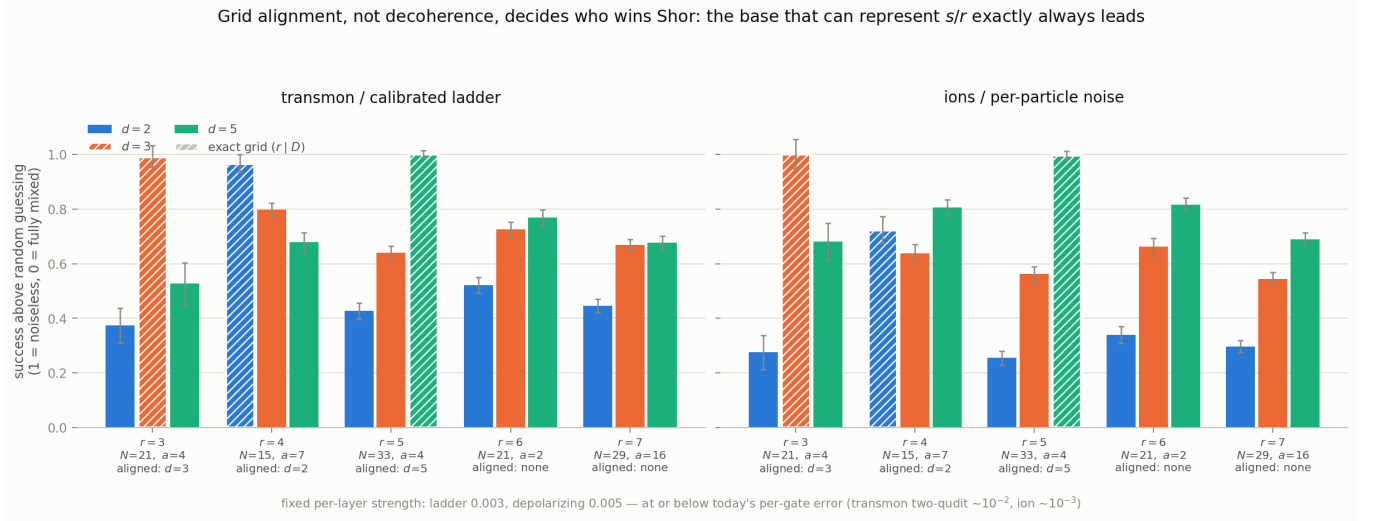


FIG. 2. Floor-corrected Shor signal for one instance per alignment class ( $r = 3, 4, 5, 6, 7$ ), under calibrated-ladder (strength 0.003) and per-particle (strength 0.005) noise, **uniform** cost model. The base for which  $r$  divides  $d^m$  wins in five of the six biased runs (three aligned instances  $\times$  two noise models); the exception is  $N = 15$  under depolarizing noise, where the aligned qubit trails the ququint. On both unbiased instances the ordering is  $d=5 > d=3 > d=2$ .

lead is physical.

*The aligned ceiling  $\varphi(r)/r$ .* Alignment interacts with the decoder, and the interaction has a closed form that follows in one line from the acceptance lemma of Appendix A. When  $r \mid D$ , the ideal distribution is supported on the  $r$  outcomes  $y = sD/r$ ; the phase  $s/r$  is an exact rational whose convergent denominators terminate at  $r/\gcd(s, r)$ , so by the lemma the decoder accepts exactly the  $s$  coprime to  $r$ :

*Corollary. For a perfectly aligned instance the noiseless decoded success probability is exactly  $\varphi(r)/r$ .*

This converts the scorability pattern of Table I from an empirical observation into a theorem. Base-2 alignment forces  $r = 2^k$ , and  $\varphi(2^k)/2^k = 1/2$  *identically*—every base-2-aligned instance has a noiseless ceiling of one half, at any modulus and any register size, which is why its floor meets its baseline and no converse base-2 control can ever be constructed ( $N = 15$  is the only qubit-aligned instance with usable dynamic range for a different reason: its floors happen to sit low). The corollary also scopes the  $\approx 0.2$  alignment price: all three aligned instances tested have *prime*  $r$ , where  $\varphi(r)/r = 1 - 1/r$  is near-maximal and the decoder rejects almost nothing; for  $r = p^k$  the gift shrinks to  $1 - 1/p$ , and the alignment price quoted above is therefore calibrated on the alignment class where it is largest. The quantity that decides the gift is grid alignment *weighted by  $\varphi(r)/r$* , not alignment alone.

*Ensemble over the multiplicative group.* Shor draws  $a$  uniformly from the units of  $\mathbb{Z}_N$ , so we also score full ensembles—every  $a \neq 1$ —at  $N = 21$  (eleven values),  $N = 33$  (nineteen), and  $N = 55$  (thirty-nine), on the demo registers; Table I scores each order class. At

TABLE I. Multiplicative-group ensembles: mean floor-corrected signal by order class for  $d = 2/3/5$ , under the calibrated ladder and per-particle depolarizing channels (**uniform** cost, marked operating points; exact density matrices at  $N = 21$ , 400 trajectories per point at  $N = 33, 55$ ). “mult.” is the class multiplicity among the units  $a \neq 1$ ; the aligned column names the base whose grid the order divides. Dashes mark unscorable classes—floor meeting or exceeding the noiseless baseline (Sec. II)—which include *every* base-2-aligned class at all three moduli.

| $N$ | class (mult.)   | aligned | ladder $d=2/3/5$ | depol. $d=2/3/5$ |
|-----|-----------------|---------|------------------|------------------|
| 21  | $r=6$ (6/11)    | —       | 0.52/0.73/0.76   | 0.33/0.67/0.79   |
|     | $r=3$ (2/11)    | 3       | —                | —                |
|     | $r=2$ (3/11)    | 2       | —                | —                |
| 33  | $r=5$ (4/19)    | 5       | 0.40/0.68/1.00   | 0.22/0.55/0.97   |
|     | $r=10$ (12/19)  | —       | 0.45/0.60/0.82   | 0.23/0.51/0.78   |
|     | $r=2$ (3/19)    | 2       | —                | —                |
| 55  | $r=5$ (4/39)    | 5       | 0.39/0.64/1.00   | 0.23/0.54/0.99   |
|     | $r=10$ (12/39)  | —       | 0.44/0.58/0.82   | 0.23/0.49/0.81   |
|     | $r=4, 2$ (7/39) | 2       | —                | —                |
|     | $r=20$ (16/39)  | —       | —                | —                |

$N = 21$  every  $r = 2$  and  $r = 3$  instance falls below the 0.15 span rule of Sec. II in every base—small orders collapse the metric, as disclosed there—so the scorable ensemble is precisely the unbiased  $r = 6$  class, where the ensemble-mean signal preserves the qudit ordering. Raw success needs no span but carries base-dependent floors (0.38/0.36/0.39 on average here): subtracting them, the full-ensemble excess is  $-0.006/+0.030/+0.019$  (ladder) and  $-0.006/+0.028/+0.025$  (depolarizing)—so the ensemble-robust statement is that both qudits clear the

TABLE II. Floor-corrected signal on the unbiased instance ( $N = 21$ ,  $r = 6$ ) under three gate-cost models, at a single common strength of 0.005 for both channels (the demo operating point; ladder figures elsewhere use the marked transmon operating point 0.003). Rightmost column: total circuit cost in time-layers for  $d = 2/3/5$ . The serial-schedule winner is decided by the cost model and the dimension, not the noise model; Sec. VIII charges measured single-qudit costs and concurrency against these cells. No transmon entangler beyond  $d = 3$  has been demonstrated, so the ladder  $d = 5$  cells are counterfactual hardware.

| noise  | cost            | $d=2$ | $d=3$        | $d=5$        | layers       |
|--------|-----------------|-------|--------------|--------------|--------------|
| depol. | <b>uniform</b>  | 0.331 | 0.667        | <b>0.782</b> | 57/26/15     |
|        | <b>ion</b>      | 0.331 | 0.497        | <b>0.502</b> | 57/44/42     |
|        | <b>pavlidis</b> | 0.331 | <b>0.461</b> | 0.361        | 57/48.5/62.2 |
| ladder | <b>uniform</b>  | 0.282 | 0.578        | <b>0.631</b> | 57/26/15     |
|        | <b>ion</b>      | 0.282 | <b>0.374</b> | 0.256        | 57/44/42     |
|        | <b>pavlidis</b> | 0.282 | <b>0.333</b> | 0.132        | 57/48.5/62.2 |

qubit, while the  $d=5 > d=3$  ordering is a property of the scorable class rather than of the full ensemble. The ensemble a Shor user actually samples thus reproduces the qudit-over-qubit conclusion.

The larger moduli add alignment gifts in both directions and close the loop quantitatively. The base-5-aligned  $r = 5$  class hands the ququint 1.00/0.97 at  $N = 33$  and 1.00/0.99 at  $N = 55$  (ladder/depolarizing), while the unbiased  $r = 10$  class gives the strict ordering  $d=5 > d=3 > d=2$  at both moduli without any gift. The ququint’s aligned-over-unaligned excess—+0.19/+0.18 at  $N = 33$ , +0.18 at  $N = 55$ —independently reproduces the  $\approx 0.2$  alignment price measured by the within-modulus control: the ensemble structure is predicted class by class. Every base-2-aligned class ( $r = 2$  at  $N = 33$ ;  $r = 2, 4$  at  $N = 55$ ) is unscorable, floors meeting or exceeding baselines—the converse collapse documented above, now observed at all three moduli. The modal class at  $N = 55$ ,  $r = 20$  (sixteen of the thirty-nine), is unscorable in *every* base, and for  $d = 2$  maximally so: at  $D = 64$  the noiseless baseline is exactly zero—the  $r = 20$  acceptance window contains no outcome at all—the decoder law of Sec. VII C deciding scorability before noise enters.

Re-running the full noise sweep on the unbiased instance (Fig. 3) reverses the study’s original conclusion: the best qudit beats the qubit at every noise strength under *all three* noise models—including the idealized ladder ( $\Gamma_k \propto k$ ,  $(\Delta\text{level})^2$  dephasing), the harshest treatment of high levels we ever used, under which the qubit now trails by 0.09–0.52. The earlier “qubits win Shor on transmons” finding was not a noise result at all; it was an arithmetic artifact, and it does not survive deconfounding under any model.

#### IV. THE COST CONDITION

Table II is the paper’s central result in one table. Its cells are exact density matrices, so they carry no statistical uncertainty; the axis that matters is the strength they are evaluated at. Sweeping it across the demo grid (0.001–0.05; `results/cost_grid_ssweep.json`) leaves the qudit-versus-qubit verdict of *every* cell unchanged wherever the metric still has dynamic range—we judge only strengths at which the best base clears 0.10 signal, beyond which all three bases sit at the random floor and the ranking is decided by third-decimal noise. That cutoff leaves live windows narrower than the grid: 0.001–0.02 for the **uniform** cells, 0.001–0.01 for depolarizing/**ion** and **pavlidis**, and only 0.001–0.005 for their ladder counterparts, so the stability claim is scoped to those windows. One piece of over-precision in the table should be read down, however: in the depolarizing/**ion** cell the two qudits are tied rather than ordered (0.502 against 0.497, and the ordering reverses to  $d = 3$  at  $s \leq 0.002$ ), so that cell establishes that a qudit wins, not which. The decomposition direction of the condition holds in a weakened, dimension-graded form. Under  $d^2$ -per-entangler decomposition (**pavlidis**; single-qudit gates cost one layer, Sec. II) the serial demo grid no longer hands the qubit a sweep: the qutrit beats the qubit on *both* channels (0.461 vs 0.331 per-particle, 0.333 vs 0.282 on the ladder; replicated at 1000-trajectory statistics in the  $d = 7$  demo grid, 0.445 vs 0.338 and 0.339 vs 0.290), the ququint splits (0.361 wins per-particle, 0.132 loses the ladder), and  $d = 7$  loses everywhere. The grading tracks the layer table. At  $d = 3$  the corrected surcharge still leaves the circuit *shorter* than the qubit’s (48.5 vs 57 layers), so decomposition trims the qutrit’s margin without surrendering either compression; at  $d = 5$  the surcharge overtakes the depth compression (62.2 vs 57) and only the width compression (5 vs 11 carriers) survives—enough under nearly-flat per-particle damage, not under the ladder’s escalating per-event damage; by  $d = 7$  neither survives. Eigenstate QPE moves with Shor (its per-particle **pavlidis** cell is now a qudit win, 0.469 against 0.396, no longer a dead heat, while its ladder cell stays with the qubit), and Grover, whose compression is halved, is where decomposition still bites hardest (Sec. VI). Two caveats keep the serial **pavlidis** qutrit wins from reading as a working no-native-gate route. On the ladder the win is an artifact of the uncharged serial schedule: it dies at single-qudit charge  $\alpha^* = 1.44$ , below any measured value, and flips to the qubit under concurrent scheduling (Sec. VIII). Under per-particle noise it is real but thin: it survives concurrency (+0.130  $\rightarrow$  +0.076) and holds up to the top of the measured single-qudit-cost range ( $\alpha^* = 3.17$  against measured 2.15–3.15). Under a native qudit gate (**uniform**) the width-and-depth compression is untouched and qudits win on both hardware classes. The intermediate linear cost (**ion**) is where dephasing level decides: on per-particle hardware the qudit advantage survives (the tied qutrit/ququint pair at

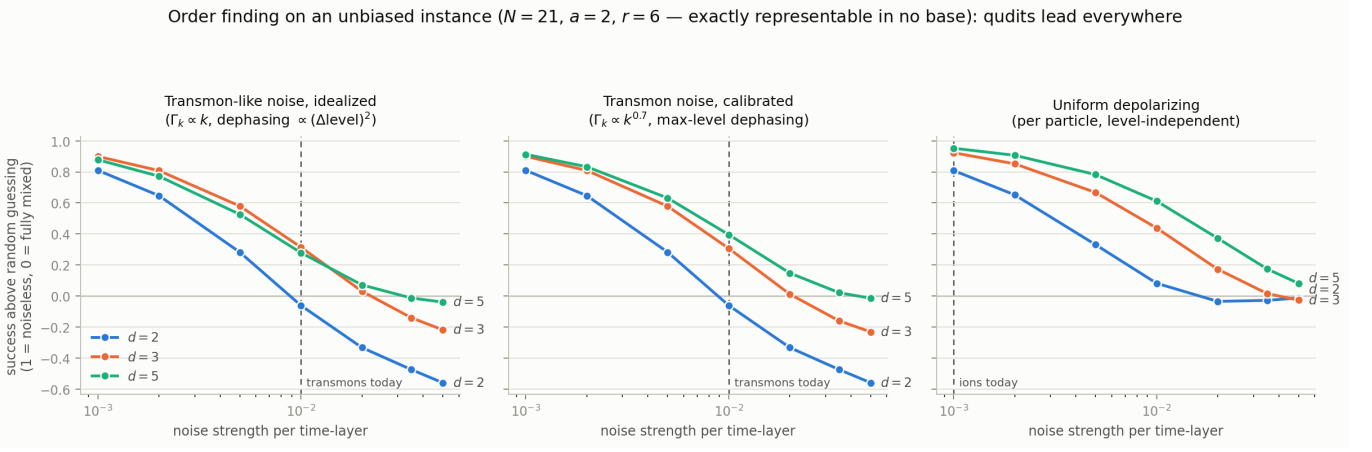


FIG. 3. Unbiased-instance noise sweep ( $N = 21$ ,  $a = 2$ ,  $r = 6$ ), exact density-matrix evolution, **uniform** cost model (native entangler). Floor-corrected signal vs noise strength for the idealized ladder, calibrated ladder, and per-particle depolarizing channels. The best qudit leads the qubit at every strength under all three models; the single exception to a uniform qudit lead is the deepest depolarizing point ( $s = 0.05$ ), where the qutrit alone dips marginally below the qubit with all signals within 0.09 of zero.

0.497/0.502 against 0.331), while at free-evolution ladder dephasing the verdict splits—the qutrit still wins Shor but the ququint loses (0.256 vs 0.282), eigenstate QPE is a dead heat (advantage +0.00), and Grover passes to the qubit for both qudits (Sec. VI). Refocusing repairs this entire column (Sec. VIII). Eigenstate QPE (never confounded, its golden-ratio target phase being irrational in every base) otherwise tracks Shor, with advantages of +0.42 (**uniform**) and +0.20 (**ion**) on per-particle noise. The decomposition direction is therefore no longer a clean forfeit: it fails outright at  $d \geq 5$  on the ladder and at  $d = 7$  everywhere, fails at  $d = 3$  once measured single-qudit costs or concurrency are charged on the ladder, and is survived—thinly—by the decomposed qutrit under per-particle noise.

*The exposure inequality.* The grid’s verdicts are readable from closed-form accounting before any simulation. Writing  $C_d$  for the carrier count,  $L_d$  for the serial layer count, and  $\Delta_d$  for the per-carrier-layer damage of the channel (0.75/1.46/2.83 per unit  $s$  at  $d = 2/3/5$  on the calibrated ladder,  $1 - 1/d^2$  per unit  $p$  under depolarizing; Sec. VII), the qudit wins a cell iff

$$C_d L_d \Delta_d < C_2 L_2 \Delta_2. \quad (2)$$

Evaluated on Table II ( $C = 11/7/5$ ,  $L$  from the layers column), Eq. (2) predicts 11 of the 12 qudit-versus-qubit verdicts and the winner in four of the six rows outright (a fifth—depolarizing/**pavlidis**—is a predicted tie at 1%, split by the decoder toward the qutrit); its one clean miss is the ladder/**pavlidis** qutrit cell—the cell the concurrency and measured single-qudit charges kill independently (Sec. VIII). What the inequality cannot supply is the *margin*: decoded success is not a function of accumulated damage, because the decoder acts on the full outcome distribution, not on the end-state fidelity it determines (Sec. VII). The distribution is what the

quantum simulation adds.

Cost model and noise model are not independent: each pairing describes a platform, and both *physically matched* pairings favor ququints—trapped-ion qudits (**ion** cost, per-particle noise) and native-cross-Kerr transmons (**uniform** cost, calibrated ladder). The ion pairing presumes shielded, refocused operation; Sec. VIII prices the unmitigated alternative, and it is severe. The pessimistic cells of the grid are mostly mismatched combinations; the genuine failure mode remains the absence of a native entangler (**pavlidis**)—fatal at  $d \geq 5$  on the ladder and under any realistic accounting at  $d = 3$  there, survivable only by the per-particle qutrit—which is precisely the regime of synthesis-based compilation [33, 36].

*Matched control dimension.* The demo grid compares  $D = 64/81/125$ , and Sec. VII C shows decoder tolerance grows with  $D$ —the ququint enters Table II with twice the qubit’s acceptance set (16 vs 8 outcomes). Equalizing  $D$  resolves the worry in the qudits’ favor: at  $d = 2$ ,  $m = 7$  ( $D = 128$ , matching the ququint’s 125) the qubit scores 0.47/0.30 (ladder/depolarizing) against 0.51/0.33 at  $D = 64$ —its extra decoder tolerance is outweighed by the deeper circuit’s added exposure—while the ququint holds 0.76/0.78 and the qutrit is stable between  $D = 81$  and 243 (0.73/0.67 vs 0.71/0.66). Unequal  $D$  biases the demo grid against the qudits, not for them.

An independent check: Janković *et al.* [34] derive, by linear response over Haar-random gates under pure dephasing, the critical gate-efficiency ratio  $(d^2 - 1)/(3 \log_2 d)$  a single qudit must clear to beat a multi-qubit register at matched Hilbert dimension—1.68 at  $d = 3$ , 3.45 at  $d = 5$ , and 5.70 at  $d = 7$ . Their criterion is stated for ratios of gate *times* in units of the decoherence time; reading our layer-count ratios as gate-efficiency ratios identifies the two, which holds when per-layer duration and coherence time scale in the same ratio across



bases—an assumption, not a derived fact, and one the structured channels of Sec. VIII can violate. Under that identification the gate-level criterion is consistent with our algorithm-level winner in four of six cost/dimension cases on the ladder channel (`results/jankovic.json`), including the tight case ( $d = 5$  **uniform** clears 3.45 at 3.80 and wins). Both misses ( $d = 3$  **ion** at ratio 1.30, and  $d = 3$  **pavlidis** at 1.18, each below the 1.68 bar yet winning) are in the conservative direction their pure-dephasing assumption predicts, our calibrated relaxation being gentler ( $\Gamma_k \propto k^{0.7}$ ); the  $d = 7$  **uniform** case below misses the same way.

*The seventh dimension.* A demo-size  $d = 7$  run on the unbiased instance (1000 trajectories per point, both channels, all three cost models; base 7 has the same mean residual misalignment, 0.300, as bases 3 and 5) extends the grid one prime higher and sharpens the condition. A caveat on the cost inputs: measured qudit entangling gates stop at  $d = 5$  [32] (Ringbauer’s are characterized at  $d = 3$  [1]), so at  $d = 7$  every cost model is an extrapolation beyond any measurement. The decomposition verdict is unambiguous at this dimension: **pavlidis** loses to the qubit in all three conditions ( $-0.05$  to  $+0.17$  signal against the qubit’s  $0.29$ – $0.49$ ), the seventh prime being where decomposition finally forfeits the advantage outright even in the serial schedule. The native-gate direction survives—under **uniform** cost the  $d = 7$  register beats the qubit on both channels ( $0.52$  vs  $0.24$  on the calibrated ladder,  $0.77$  vs  $0.33$  under depolarizing, at  $s = 0.005$ )—but on the ladder the ordering is no longer monotone in  $d$ : the optimum sits at  $d = 5$  ( $0.64$ ), the larger per-event damage of the seventh level pulling  $d = 7$  back to  $0.52$ , while under depolarizing  $d = 7$  ( $0.772 \pm 0.018$ ) is statistically tied with  $d = 5$  ( $0.783 \pm 0.015$ ; a  $0.5\sigma$  margin) and above the rest. The intermediate **ion** cost, already split at  $d = 5$ , now fails cleanly on the ladder ( $0.05$  vs  $0.25$ ) and retains only a  $+0.05$  ( $1.6\sigma$ ) edge under depolarizing: the window between native and decomposed cost narrows with every added level, exactly as the rising break-even curve suggests. The criterion itself is conservative here—its  $d = 7$  bar of  $5.70$  is not cleared (the uniform layer ratio is  $3.80$ ), yet the  $d = 7$  register wins.

The  $d = 7$  cells enter with a control-dimension gift the rest of the grid does not get: at  $D = 343$  the acceptance set for  $r = 6$  holds 48 outcomes against the qubit’s 8 at  $D = 64$  and the ququint’s 16 at  $D = 125$ —the confound the matched- $D$  control above exists to price, six-fold at this dimension. Extending that control (`results/d7_matched_D.json`): qubit registers bracketing  $D = 343$ , at  $m = 8$  ( $D = 256$ ,  $|\mathcal{A}| = 36$ ) and  $m = 9$  ( $D = 512$ ,  $|\mathcal{A}| = 74$ ), score *lower* than at  $D = 64$ — $0.27/0.22$  under depolarizing and  $0.21/0.16$  on the ladder at the same  $s = 0.005$  (**uniform** cost, 1000 trajectories, errors  $\pm 0.02$ )—the added decoder tolerance again losing to the deeper circuit’s added exposure. Matching  $D$  therefore *widens* every  $d = 7$ -vs-qubit gap quoted above rather than closing it, and the one  $d = 7$  loss, to  $d = 5$

on the ladder, happens *despite* a  $3\times$  acceptance-set advantage at identical carrier and layer count—the seventh level’s larger per-event damage overcomes a decoder gift.

*The primes past the bar.* At  $d = 11$  the demo register hits a step:  $m = w = 2$  suffices ( $D = 121$ ), leaving four carriers and nine **uniform** layers—36 carrier-layers of exposure against the qubit’s 627—while the ladder’s per-event damage keeps climbing,  $6.8s$  and  $8.1s$  per carrier-layer at  $d = 11, 13$  against  $2.8s$  at  $d = 5$ . The two effects collide, and exposure wins (`results/d11_demo.json`; 1000 trajectories, both bases at residual misalignment  $0.300$ , matching bases 3–7): with a native gate the eleventh and thirteenth primes beat the qubit decisively on both channels ( $0.89/0.86$  vs  $0.33$  under depolarizing,  $0.68/0.67$  vs  $0.24$  on the ladder at  $s = 0.005$ ). The **ion** charge, whose  $d-1$  multiplier now costs 54–64 layers, collapses on the ladder ( $0.07/0.03$  vs  $0.25$ ) while keeping a thin depolarizing edge ( $0.40/0.45$  vs  $0.33$ ), and **pavlidis** loses everywhere ( $-0.03$  to  $+0.10$  against the qubit’s  $0.29$ – $0.34$ ). Three cautions temper the native-gate cells. The ordering in  $d$  is set by register granularity—the drop from  $m = 3$  to  $m = 2$  at  $d \geq 9$  is why  $d = 11$  scores above  $d = 7$ —so these are demo-size statements, not asymptotics. The larger  $D$  carries a larger acceptance set (floors  $14/121$  and  $24/169$ , floor-corrected as everywhere). And no entangling gate beyond  $d = 5$  exists on any platform, so every cost model here is counterfactual twice over. The gate-level break-even bar  $(d^2-1)/(3\log_2 d) = 11.6$  and  $15.1$  is not remotely cleared by the layer ratio  $57/9 = 6.3$ , yet the native register wins—the criterion errs conservative at its widest margin yet. The structural statement survives untouched: native gates keep the advantage alive at every prime tested; decomposition has forfeited it from  $d = 7$  on, and linear cost forfeits the ladder from  $d = 11$  on.

## V. SCALING WITH PROBLEM SIZE

Figure 4 sweeps precision from 6 to 14.3 bits (control dimensions 64 to 19683; the largest register is the qutrit’s sixth size,  $m = 9$ , at Hilbert-space dimension  $5.3 \times 10^5$ ) at fixed per-layer strength. Both qudits stay above the qubit at every precision on the measured overlap in all four regimes, and the gap over the qutrit widens with size under ladder noise because the qubit decays several times faster per precision bit—a ratio of  $3.6 \pm 1.0$ : slopes  $-0.049 \pm 0.005/\text{bit}$  ( $d=2$ ,  $R^2 = 0.97$ ,  $n = 4$  sizes) against  $-0.013 \pm 0.003/\text{bit}$  ( $d=3$ ,  $R^2 = 0.73$ ,  $n = 6$ ) under the calibrated ladder, while the ququint decays at a rate comparable to the qubit’s ( $-0.043 \pm 0.006/\text{bit}$ ,  $n = 3$  points:  $0.750, 0.669, 0.535$  at  $7.0, 9.3, 11.6$  bits) from a higher starting point. Slopes are weighted least squares, the per-point standard errors serving as weights because Sec. XI measures them to be calibrated; unweighted fits give the same slopes within errors but take their uncertainties from residual scatter, which at  $n = 3$  is one degree of freedom and not usable. Two readings deserve

Shor scaling with the grid-alignment confound removed ( $N = 21$ ,  $a = 2$ ,  $r = 6$ ): the qudit lead holds at every size

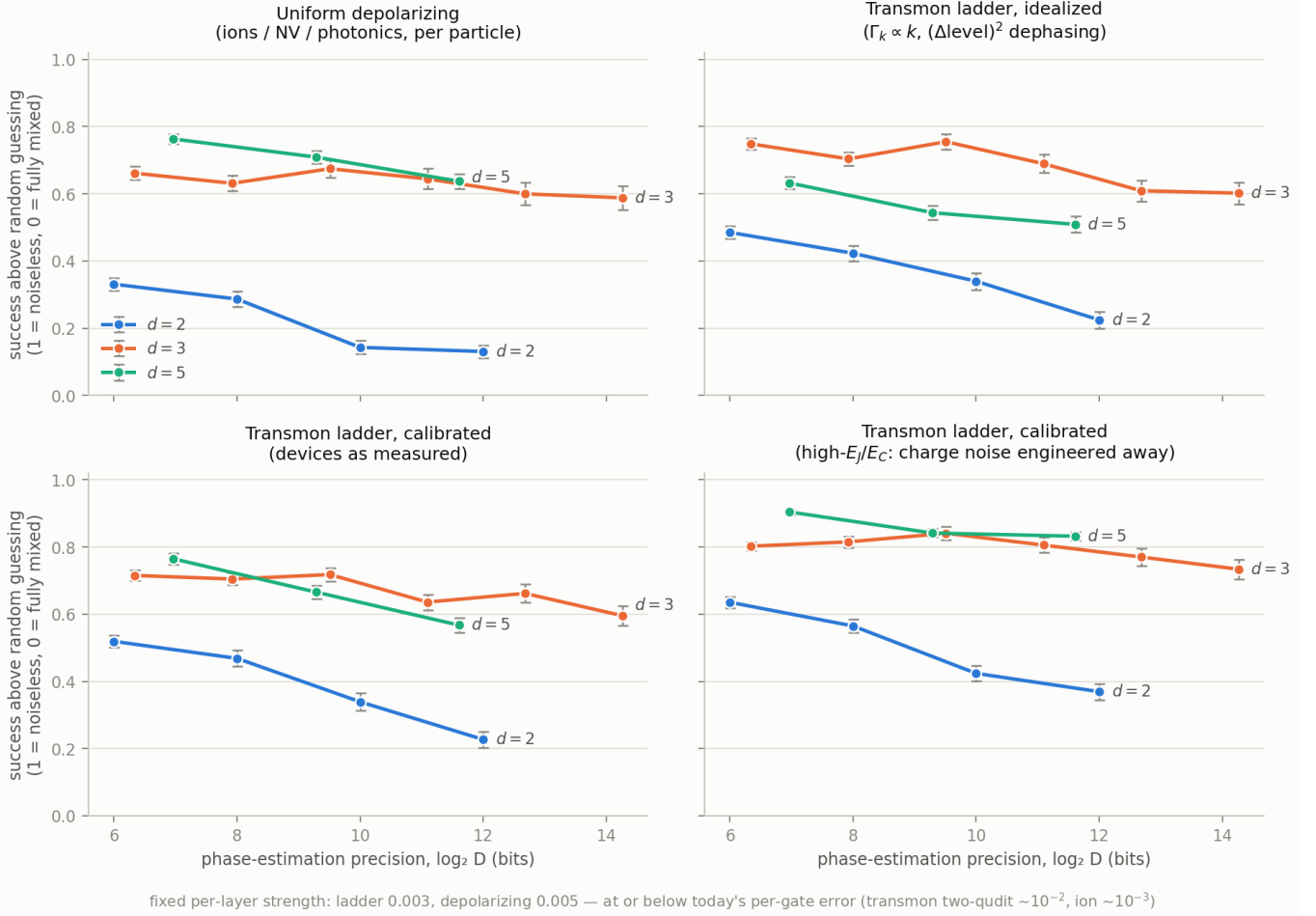


FIG. 4. Register-size scaling on the unbiased instance: floor-corrected signal vs phase-estimation precision (bits), four noise regimes (ladder regimes at strength 0.003, depolarizing at 0.005), **uniform** cost model, registers to Hilbert-space dimension  $5.3 \times 10^5$ , quantum trajectories (1000 per point).

emphasis. First, the qudit's size dependence is the shallowest in every regime, but it is flat in none of them: the family holds a plateau and then falls, and a single slope is therefore a summary rather than a model. Under the calibrated ladder the first three sizes agree within 0.013 (0.705–0.718 from 6.3 to 9.5 bits) and the whole decline is carried by the last three, the 14.3-bit point sitting  $3.5\sigma$  below the 9.5-bit one. Because both statements are spent on the same error bars, we characterized the trajectory estimator rather than assuming it (Sec. XI): across 24 independent 1000-trajectory replicas at five grid points the empirical standard error of the mean agrees with the quoted one to a ratio of 0.995 (range 0.84–1.17; no point departs at  $p < 0.2$ ), so the bars are calibrated, not conservative. A constant fit to the plateau gives  $\chi^2/\text{dof} = 0.15$  ( $p = 0.14$ ), which is unremarkable; we quote the raw spread there and reserve significance for the fall. An ear-

lier release of these sweeps returned  $\chi^2/\text{dof} = 0.015$  on the same three points, a 1.5% low-tail value that might be read as over-conservative bars. It was not: regenerating every sweep from reproducible seeds (Sec. XI) returns the ordinary value above, which is what a fluctuation does and what a mis-sized error model would not. The same shape appears under depolarizing, where four sizes hold near 0.65 before falling to  $0.59 \pm 0.04$  at 14.3 bits, a  $2.0\sigma$  drop from 9.5 bits that turns a slope consistent with zero over the first five sizes into  $-0.007 \pm 0.004/\text{bit}$  over six. The onset is later and the fall gentler in the channel whose per-event damage is flatter in  $d$ , which is the ordering Sec. VII predicts: the decoder tolerance of Sec. VII C postpones the decay of decoded success but does not outrun it. We report the plateau as an observation with a mechanism, not a law: five sizes would have read as a flat depolarizing family, and it is the sixth that resolves

the shape. One objection must be met first:  $D = d^m$  changes across the sweep, so the plateau could be residual grid misalignment drifting with  $m$ —the confound of Sec. III returning in a new form. It is not. Residual misalignment is *exactly constant* across the entire sweep for all three bases, on both benchmark instances (0.267 for  $d = 2$  and 0.300 for  $d = 3, 5$  at  $N = 21$ ; 0.286 for every base at  $N = 29$ ; constant to numerical precision, spread  $< 10^{-11}$ , over the sizes), because  $d^m \bmod r$  is periodic in  $m$  and the multiset of phase offsets it induces therefore repeats. Alignment does not vary with register size here, so it can explain neither a size-dependent nor a size-independent signal. On the fitted curves the qutrit overtakes the ququint at 8.6 bits (68% CI 8.1–9.1, calibrated ladder) and 12.5 bits (68% CI 11.3–14.7, depolarizing—an extrapolation, the ququint’s largest simulated size being 11.6 bits, though the interval’s lower end falls just inside it), and from the smallest common size under the idealized ladder—the story is not monotonic in  $d$ , and the best base depends on the target precision. Second, for eigenstate QPE the qudit advantage is decisive and grows with size: at 11.6 bits the ququint retains  $2.7\times$  the qubit’s signal as measured ( $+0.42 \pm 0.02$ ) and the margin reaches  $+0.46 \pm 0.02$  ( $2.1\times$ ) in the high- $E_J/E_C$  regime. Quoted errors are statistical; the dominant uncertainty is systematic—the spread across channel and cost-model choices documented above. Since eigenstate QPE is the quantum-chemistry workhorse, this is the result with the clearest practical consequence.

*Instance robustness.* Re-running the identical Shor sweep on the second unbiased instance ( $N = 29$ ,  $a = 16$ ,  $r = 7$ —the exactly alignment-neutral benchmark of Sec. III, whose work register is one digit wider for  $d = 3, 5$ ; 400 trajectories per point) replicates every claim above. Both qudits again clear the qubit at every size in all four regimes, and the qubit again decays fastest ( $-0.042 \pm 0.006/\text{bit}$  under the calibrated ladder, matching  $-0.049 \pm 0.005$  at  $N = 21$  within  $1.0\sigma$ ; the ququint matches at  $-0.058 \pm 0.009$  vs  $-0.043 \pm 0.006$ ,  $1.4\sigma$ ). Slopes are compared over *matched* size ranges: the  $N = 29$  sweep reaches four qutrit sizes and the  $N = 21$  sweep six, and setting a four-size slope against a six-size one that includes the fall is not a replication test. Restricted to the common range, the qutrit’s slope agrees between the instances under the calibrated ladder ( $-0.028 \pm 0.008/\text{bit}$  on  $N = 29$  against  $-0.012 \pm 0.006$  at  $N = 21$ ,  $1.7\sigma$ ) but *not* under depolarizing ( $-0.039 \pm 0.008$  against  $-0.001 \pm 0.007$ ,  $3.7\sigma$ ), where the  $N = 29$  qutrit decays while the  $N = 21$  qutrit is flat across the same four sizes. The qutrit remains the shallowest family in every regime on both instances, which is the ordering claim; its *rate* is instance-dependent in the flatter channel, and we do not claim otherwise. The  $N = 29$  sweep also stops at four qutrit sizes, so it does not reach the register depth at which the plateau above gives way. What is instance-robust is the ordering, the qubit’s fastest decay, and the qutrit-over-ququint crossing under the calibrated ladder, which recurs at 10.5 bits on  $N = 29$  (8.6 at  $N = 21$ ).

Under depolarizing the crossing is not instance-robust: on  $N = 29$  the qutrit decays *faster* than the ququint—the slope non-replication above—so no crossing occurs at any size.

## VI. GROVER SEARCH: DECOMPOSING THE MECHANISM

Everything above measures one algorithm family—the interpolation experiment of Sec. VII shows Shor *is* phase estimation on an eigenstate superposition. Grover search is the second family, chosen to attack our own weakest claim, with the prediction registered in advance (`grover_study.py` docstring, commit 6540a97, 2026-08-10): *qudits win, by less than in phase estimation*. The iteration count  $\lfloor (\pi/4)\sqrt{M} \rfloor$  is base-independent, so Grover holds the oracle count fixed across bases; its residual depth compression comes only from the narrower multi-qudit decompositions, and at matched size its total exposure compression is roughly half of Shor’s ( $5.7\times$  vs  $10.9\times$  from  $d = 2$  to  $d = 5$ ). This is a partial rather than a clean separation of width from depth, but it halves the compression while holding algorithm structure fixed. A null result would have meant the advantage was depth all along.

The prediction holds (Fig. 5): the ordering  $d = 5 > d = 3 > d = 2$  at every matched size in both noise models, with the Grover advantage at 0.33–0.50 of Shor’s. Halving the exposure compression thus costs one-half to two-thirds of the advantage: the compressed register is sufficient for a qudit advantage, and the advantage responds at least proportionally to compression. Grover also serves as an independent methodological control: it has no order, no continued fractions, and an exact  $1/M$  floor, so grid alignment is structurally impossible—its agreement rules out the continued-fraction metric as the source of the effect. The cost condition binds more tightly here than for Shor: qudits win under **uniform**, while under **pavlidis** the ququint loses at every matched size in both channels and the qutrit is competitive only below  $\approx 6$  bits (from  $\approx 6$  bits the ladder is the qubit’s at matched size, and the per-particle qutrit no better than tied)—Grover keeps none of the decomposed-qutrit advantage that Shor retains (Sec. IV). Under **ion** cost at free-evolution ladder dephasing the ququint loses Grover at every matched size and the qutrit above  $\approx 6.5$  bits (below which it retains a  $\sim 3\sigma$  edge) while the qutrit still wins Shor—the cell where the three algorithms part ways, consistent with Grover’s halved compression buying less headroom against the same gate surcharge, and the cell that refocusing repairs (Sec. VIII).

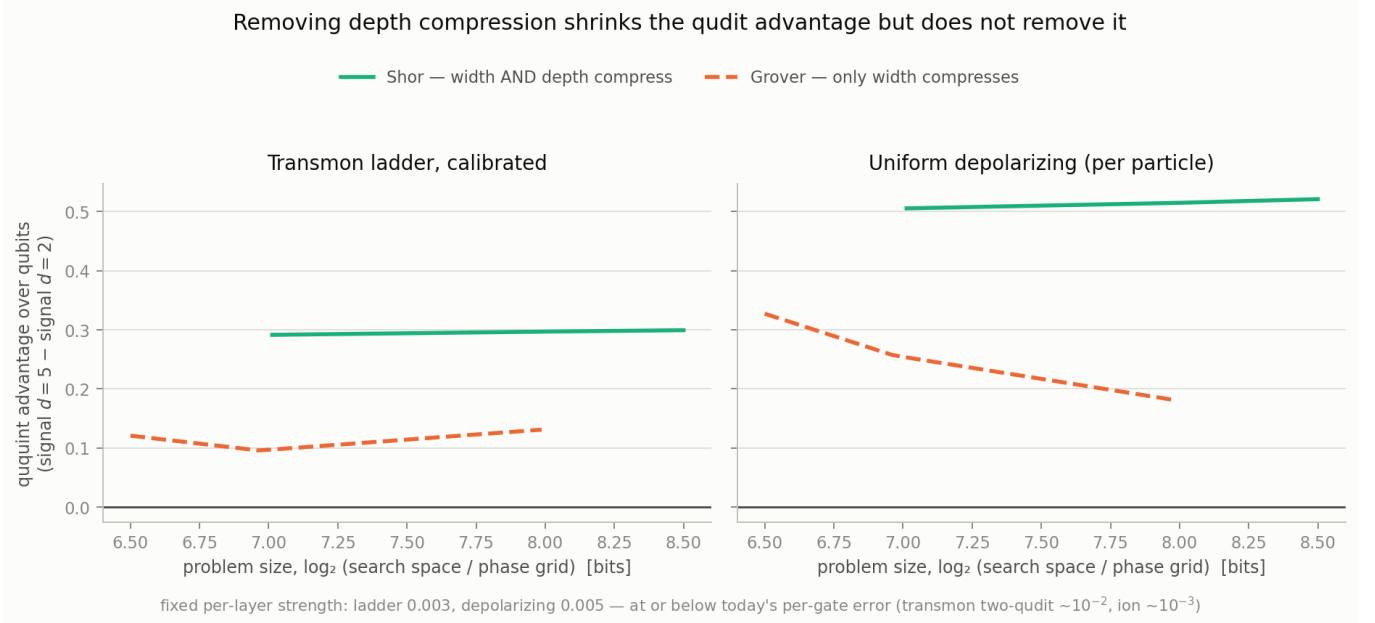


FIG. 5. Grover vs Shor floor-corrected signal at matched problem size, both noise models, **uniform** cost model (native entangler). The qudit ordering  $d=5 > d=3 > d=2$  holds at every size in both models; the Grover advantage is 0.33–0.50 of Shor’s, consistent with its halved exposure compression.

## VII. THE MECHANISM: DAMAGE UNITS AND THE DECODER

### A. Entanglement: a falsified hypothesis

Shor entangles control and work registers; eigenstate QPE does not. An early hypothesis attributed the (confounded) Shor/QPE difference to which-path dephasing through the work register. Because Shor’s work input  $|x=1\rangle$  is an equal superposition of the  $r$  eigenstates of the modular multiplier, QPE on a  $K$ -fold eigenstate superposition interpolates continuously from eigenstate QPE ( $K=1$ ) to the Shor regime ( $K=r$ ) with circuit, metric, noise, and cost held fixed; the control–work entanglement entropy is exactly  $\log_2 K$  bits (the sweep reaches  $K=4$ , i.e. 2.0 of the Shor regime’s  $\log_2 6 \approx 2.6$  bits). The measured effect—the slope of the quint advantage in entanglement bits—is far too small, where it exists at all: consistent with zero under native-gate cost ( $+0.002 \pm 0.012$  per bit on the calibrated ladder,  $-0.011 \pm 0.011$  under depolarizing) and  $-0.032$  to  $-0.043$  ( $\pm 0.013$ ) under ion cost, bounding the effect over the full two-bit range below  $\approx 0.09$  signal against the  $\approx 0.5$  gap—measured on the then-current, confounded  $N=15$  comparison—it was meant to explain (that gap itself did not survive the de-confounding of Sec. III). We report it as a falsified mechanism—and it was chasing this failure that exposed the grid-alignment confound of Sec. III.

### B. Exposure in damage units, and the decoder boundary

The surviving mechanism candidate is total noise exposure: carriers  $\times$  time-layers. If it were sufficient, signal would be a function of exposure  $\times$  strength alone and all points—both algorithms, all bases, all sizes—would collapse onto one curve. They do not ( $R^2 = 0.67/0.77$  for ladder/depolarizing), and the failure decomposes into two identifiable pieces.

*Piece 1: exposure must be counted in damage, not events.* One noise layer does  $d$ -dependent harm. Writing  $S$  for the one-layer superoperator, the per-carrier-layer entanglement infidelity is  $1 - F_e$  with  $F_e = \text{tr } S/d^2$ : under the calibrated ladder at strength  $s$  it is  $0.75s$ ,  $1.46s$ , and  $2.82s$  for  $d=2, 3, 5$ —a quint takes  $\sim 4\times$  a qubit’s damage per event—and  $p(1 - 1/d^2)$  under depolarizing. With the abscissa exposure  $\times (1 - F_e)$ , Grover’s three bases collapse onto a single exponential: per-family decay rates  $0.44/0.49/0.43$  (a  $1.1\times$  spread, against  $3.6\times$  in event units), each family log-linear with  $R^2 \geq 0.996$ . The reweighting is a ladder effect: depolarizing damage is already nearly flat in  $d$  ( $0.75p$ – $0.96p$ ), and there the signal rows of Table III gain nothing from it. The unit itself carries a convention:  $1 - F_e$  tends to  $1 - 1/d^2$  for a fully depolarizing channel, growing with  $d$  by construction, and the  $d$ -normalized alternative—the average gate infidelity  $1 - F_{\text{avg}} = [d/(d+1)](1 - F_e)$ —rescales each family’s rate by  $(d+1)/d$ . In those units (**results/favg\_rescore.json**) the ladder spread widens back to  $1.29\times$  while the depolarizing spread *tightens* from  $1.52\times$  to  $1.23\times$ : the two channels move in opposite di-



TABLE III. The collapse test: goodness of a single two-parameter exponential  $Ae^{-kx}$  pooled over both algorithms, all bases and sizes.

| ordinate | abscissa                   | ladder $R^2$ | depol. $R^2$ |
|----------|----------------------------|--------------|--------------|
| signal   | exposure $\times$ strength | 0.67         | 0.77         |
| signal   | exposure $\times$ damage   | 0.84         | 0.77         |
| fidelity | exposure $\times$ damage   | <b>0.97</b>  | <b>0.99</b>  |

rections, so neither unit is uniformly flattering, and the  $1.1\times$  figure should be read as convention-dependent at the  $\pm 25\%$  level. The collapse’s headline row is not: the pooled fidelity fits of Table III are unchanged to two decimals in either unit, marginally *improving* in  $F_{\text{avg}}$ .

*Piece 2: the residual is the decoder, not the noise.* With units fixed, a nested fit finds one shared amplitude with a per-*algorithm* decay rate at  $R^2 = 0.953$  on the ladder and 0.93–0.94 under depolarizing, the range spanning the event- and damage-unit abscissae (`results/exposure_collapse.json`, nested fits). The split is stark at the family level: Grover is a textbook exponential in damage, while Shor’s  $d = 3$  family holds a flat signal of  $\approx 0.73$  while its exposure triples. A candidate explanation is that Grover’s “decoder” is the identity—its signal *is* marked-state survival—whereas Shor’s signal passes through continued-fraction order recovery, whose error tolerance grows with register size: added digits add exposure but also add redundancy the decoder exploits.

This is directly testable: if the decoder story is right, end-state *fidelity*  $\langle \psi_{\text{ideal}} | \rho | \psi_{\text{ideal}} \rangle$  should obey the one-curve law that signal does not. It does (Table III): one shared exponential in damage units fits the fidelity of both algorithms at  $R^2 = 0.97$  (ladder) and 0.99 (depolarizing), with fitted amplitude  $\approx 1$  as the zero-noise limit demands; granting the algorithms separate rates adds only 0.02/0.003. The decoder gap is visible in the raw numbers: Grover’s fidelity equals its signal to better than 0.006 under the ladder and 0.025 under depolarizing at every point, while Shor’s  $d = 3$  family loses two thirds of its fidelity ( $0.55 \rightarrow 0.18$ ) with signal flat, and at  $d = 2$ ,  $m = 12$  under depolarizing, continued fractions decode a signal of  $0.131 \pm 0.018$  from a state with fidelity  $6.4(2.0) \times 10^{-4}$ . The mechanism claim, in the form the data support: *accumulated channel damage is the law for the quantum state; the residual algorithm dependence of decoded success sits in the decoder*—for which Sec. VII C gives an exact quantitative account, derived and verified against the decoder itself.

*What the collapse does and does not prove.* For a product of near-identity incoherent channels, log-fidelity is additive in the per-application entanglement infidelity to first order, so a single exponential with amplitude  $\approx 1$  is the *null expectation*, not a discovery. The informative content of Table III is therefore bounded as well as measured. In linear fidelity the shared fit gives

$R^2 = 0.97/0.99$ ; rescored in *log* fidelity—the metric that weights the deep tail—the same fit gives  $R^2 = 0.74\text{--}0.75$  (0.87–0.93 if refit there). The deepest Shor points, whose first-pass estimates carried 40–48% statistical error at 100–200 trajectories, were re-measured at 1600 trajectories each (`results/collapse_tail_deep.json`); the re-measurement moved the deepest depolarizing point from  $2.7$  to  $6.4(2.0) \times 10^{-4}$ —the 100-trajectory mean of this heavy-tailed estimator had been skewed low—and tightened the other two. Against the shared law the deep endpoints now read as a test rather than a resolution limit: the law holds to a factor of 1.12 down to fidelity  $2.1(3) \times 10^{-2}$  on the ladder and, excepting a single floor-pinned Grover point, to a factor of 1.5 at  $6.4(2.0) \times 10^{-4}$  under depolarizing; every departure is a Grover point approaching its  $1/\dim \mathcal{H}$  fidelity floor (the deepest, at  $5.3 \times 10^{-3}$ , sits  $1.4\times$  above its floor of  $1/256$  and  $82\times$  above the fit)—a floor that varies across exactly the bases and sizes the collapse unifies. What survives everywhere is the decoder residual, which no channel-level argument predicts. Conversely, the law’s failure mode is error that does not compose incoherently: correlated or coherent noise breaks the additivity, and that is precisely what the deep hardware circuit exhibits (Sec. IX)—which is why the law is stated for Markovian incoherent channels only (see Limitations).

### C. Why the decoder gains tolerance with size

The size dependence of Shor’s decoder tolerance has a closed form, and because the decoder is classical we can measure it exactly rather than argue it. Continued-fraction recovery succeeds for any outcome  $y$  whose phase  $y/D$  lies within  $1/(2r^2)$  of some  $s/r$ —the classical convergent guarantee [25, 37]. This sufficient condition gives an acceptance window around each of the  $r - 1$  peaks spanning  $1/r^2$  in phase and holding  $\sim D/r^2$  outcomes, while the noiseless interference peak stays  $\sim 1$  outcome wide at every size. The window therefore grows linearly in  $D = d^m$ , exponentially in register size, whereas noise-induced broadening grows only polynomially in  $m$  (exposure = carriers  $\times$  layers). The decoder’s redundancy therefore outpaces the added exposure over a range of sizes, which is why Shor’s signal families hold a plateau while their fidelity decays exponentially—and why Grover, whose acceptance set is the single marked item at every size, shows no such plateau. The reprieve is finite: the acceptance set grows linearly in  $D$  but the accepted outcomes must still carry amplitude, and once fidelity has fallen far enough the decoded signal follows, which is the fall the sixth qutrit size exposes in Sec. V. Decoder tolerance postpones the decay of decoded success; it does not repeal it.

Table IV runs the project’s own continued-fraction routine over every outcome  $y \in [0, D)$  and counts the acceptance set exactly; the growth from 1.6 accepted outcomes at  $D = 64$  to 116 at  $D = 4096$  is a  $73\times$  gain in tolerance

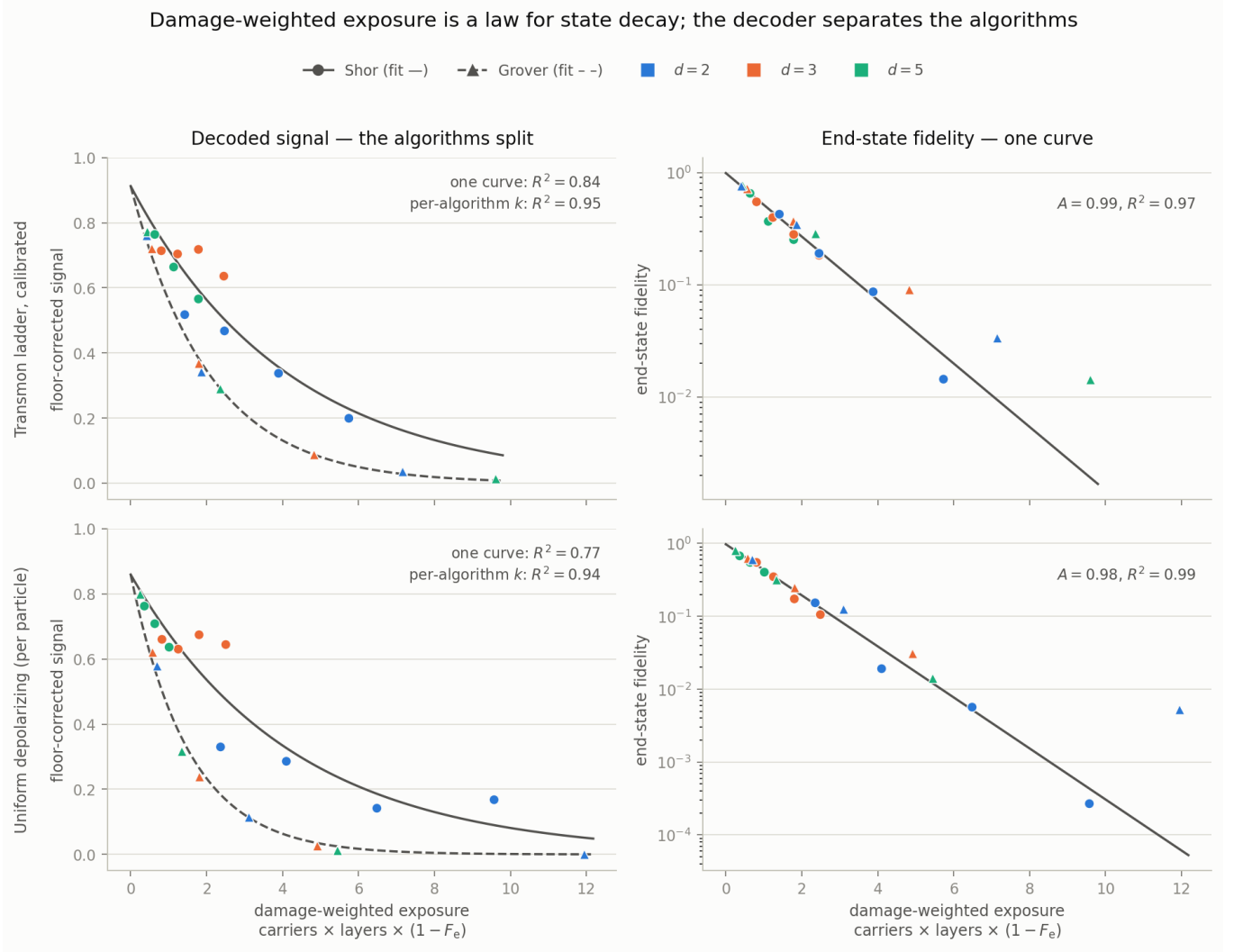


FIG. 6. Damage units and the decoder. Left column: floor-corrected decoded signal against damage-weighted exposure—the algorithms refuse a common curve ( $R^2 = 0.77$ – $0.84$ ), and granting a per-algorithm decay rate (solid: Shor; dashed: Grover) recovers  $R^2 = 0.93$ – $0.95$ ; Shor’s  $d = 3$  family (orange circles) sits nearly flat while its exposure triples. Right column: end-state fidelity on the same abscissa (log scale) falls on a single exponential with amplitude  $\approx 1$  for both algorithms, all bases and sizes ( $R^2 = 0.97/0.99$ ; **uniform** cost model). The residual algorithm structure in decoded signal is therefore the classical decoder, not unexplained noise physics.

against a peak that never widens. The acceptance set moreover has an exact description. The decoder accepts  $y$  if and only if some convergent denominator  $q$  of  $y/D$  is a multiple of  $r$  with  $q \leq N$  (Appendix A proves the equivalence and reconciles it with the divisor-form recovery of the standard analyses), which reduces  $A$  to pure number theory: for a reduced fraction  $p/q$ , the phases with  $p/q$  among their convergents form the open interval between the Stern–Brocot mediant  $(p + p')/(q + q')$  and  $(2p - p')/(2q - q')$ , where  $p'/q'$  is the penultimate convergent of  $p/q$ —a classical fact of continued-fraction theory [37, 38]. A single phase can carry *several* admissible convergent denominators (at  $N=55, r=5$  the chain 5, 10, 25 is realizable,  $25 = 2 \cdot 10 + 5$ ), so the intervals for different admissible  $q$  overlap and cannot simply be

summed. Partitioning  $A$  by the *first* admissible convergent of  $y/D$  is disjoint by construction, and restricting each interval to the phases whose earlier convergents contain no admissible denominator gives the exact finite- $D$  law: writing  $\chi(p/q) = 1$  when no convergent of  $p/q$  before  $q$  has denominator divisible by  $r$ ,

$$|A| = \sum_{k=1}^{\lfloor N/r \rfloor} \sum_{\substack{0 < p < kr \\ \gcd(p, kr)=1 \\ \chi(p/kr)=1}} \left| D \cdot I\left(\frac{p}{kr}\right) \cap \mathbb{Z} \right|, \quad (3)$$

with  $I(p/q)$  the open mediant interval above. Equation (3) reproduces the enumerated  $|A|$  *without error*—outcome for outcome, on all 27 instance/size combinations tested (the 11 register sizes of Table IV plus 16

TABLE IV. The decoder measured, not estimated: exact size of the continued-fraction acceptance set  $A = \{y \in [0, D) : \text{decode}(y) = r\}$  on the benchmark instance ( $N = 21, r = 6$ ), obtained by running the decoder over every outcome. “per peak” is  $|A|/(r - 1)$ ; the envelope estimate is  $D/r^2$ ; “law” is the *asymptotic* totient sum of Eq. (5) per peak, whose small deviations from the measured column are the small-denominator asymptote error quantified in the text—the exact finite- $D$  formula, Eq. (3), reproduces the  $|A|$  column without error at every size. No simulation is involved.

| $d$ | $m$ | $D$  | $ A $ | per peak | $D/r^2$ | law   |
|-----|-----|------|-------|----------|---------|-------|
| 2   | 6   | 64   | 8     | 1.6      | 1.8     | 1.8   |
| 2   | 8   | 256  | 36    | 7.2      | 7.1     | 7.2   |
| 2   | 10  | 1024 | 148   | 29.6     | 28.4    | 28.9  |
| 2   | 12  | 4096 | 582   | 116.4    | 113.8   | 115.7 |
| 3   | 4   | 81   | 10    | 2.0      | 2.2     | 2.3   |
| 3   | 5   | 243  | 36    | 7.2      | 6.8     | 6.9   |
| 3   | 6   | 729  | 100   | 20.0     | 20.2    | 20.6  |
| 3   | 7   | 2187 | 308   | 61.6     | 60.8    | 61.8  |
| 5   | 3   | 125  | 16    | 3.2      | 3.5     | 3.5   |
| 5   | 4   | 625  | 88    | 17.6     | 17.4    | 17.6  |
| 5   | 5   | 3125 | 442   | 88.4     | 86.8    | 88.2  |

within-modulus (order,  $D$ ) pairs). As  $D \rightarrow \infty$  each count becomes  $D$  times an interval measure; the exact per-denominator measure is

$$\mu(q) = \frac{2}{q} \sum_{\substack{0 < u < q \\ \gcd(u, q) = 1}} \frac{1}{q + u}, \quad (4)$$

and summing it over the admissible denominators,  $|A|/D \rightarrow \sum_k \mu(kr)$  minus the first-admissible overlap corrections. The overlaps vanish on five of our six instances— $r$  and  $2r$  are never convergent denominators of the same phase (if  $q_j = r$  then  $q_{j+1} = ar + q_{j-1}$  with  $0 < q_{j-1} < r$  is not a multiple of  $r$ , and  $q_{j+2} > 2r$ ), and deeper coincidences require the chain structure above—so there the bare  $\mu$ -sum matches the enumerated limit to better than  $5 \times 10^{-5}$  relative; on the sixth ( $N = 55, r = 5$ , eleven admissible denominators) the overlaps contribute +3.5%. Replacing  $\mu(q)$  by its large- $q$  asymptote  $2 \ln 2 \varphi(q)/q^2$  gives the totient form

$$\frac{|A|}{D} \approx 2 \ln 2 \sum_{k=1}^{\lfloor N/r \rfloor} \frac{\varphi(kr)}{(kr)^2}, \quad (5)$$

with  $\varphi$  Euler’s totient (summed over *all* denominators  $q \leq Q$  the weight  $\mu$  reproduces the almost-everywhere count  $(12 \ln 2 / \pi^2) \ln Q$  of convergent denominators below  $Q$  [38], so the measure itself is classical—the content of Eqs. (3)–(5) is its restriction to the denominators the decoder admits). The residual of Eq. (5) is dominated by the asymptote substitution, not by finite-size discreteness: at the small denominators that carry the

sum,  $2 \ln 2 \varphi(q)/q^2$  errs by  $\pm 1$ –2% in both directions, and the measured deviations—between  $-0.8\%$  and  $+0.7\%$  on five instances and  $+4.1\%$  on the sixth, constant in  $D$  over three decades—decompose accordingly (asymptote error on the five overlap-free instances; asymptote plus overlap on the sixth). The law settles both scaling directions at once. In  $D$  it is exactly linear—the measured  $|A| \propto D^{1.03 \pm 0.01}$  ( $R^2 = 0.999$ ) carries its small excess over slope 1 purely from window discreteness. In  $r$  it replaces the  $1/r^2$  envelope: comparing  $r = 5$  with  $r = 10$  on identical registers (the within-modulus pairs of Sec. III, swept to  $D \geq 5r^2$  so the window exceeds one outcome) gives measured per-peak ratios of 9.6 ( $N = 33$ ) and 8.9 ( $N = 55$ ) where  $1/r^2$  predicts 4.0; the law’s totient-sum ratios are 9.7 and 9.3. It also explains the envelope’s apparent success in Table IV as an accident of the benchmark instance: at  $r = 6, N = 21$  the naive  $(r - 1)/r^2 = 0.139$  sits within 2% of the law’s 0.141 because two errors compensate—the true  $q = r$  window measure  $2 \ln 2 \varphi(r)/r^2$  is only  $0.55\times$  the envelope’s, and the admissible multiples  $2r, \dots, \lfloor N/r \rfloor r$  restore the difference. On the  $r = 5$  instances, where the cancellation fails, the envelope is off by  $2.5$ – $3\times$  while the law stays within 4.2%.

*Relation to success-probability analyses of order finding.* An exact literature bounds the success of continued-fraction post-processing on base-2 registers: Shor’s original  $4/\pi^2$  asymptotics [25], the sharp divisor-recovery bounds of Gerjuoy [39] and of Bourdon and Williams [40], Ekerå’s proof that a single run recovers  $r$  with probability approaching one [41], the tight two-sided window bounds of Magdon-Ismael and Dong [42], the self-contained treatise of Barzen and Leymann [43], and, for eigenstate phase estimation, Chappell *et al.* [44]. These analyses score outcomes inside a tolerance window of the peaks via the convergent guarantee—a sufficient condition—and typically certify recovery of a divisor  $r/\gcd(s, r)$  that a classical search then lifts to  $r$ . The law above differs in kind: it is an exact, outcome-for-outcome description of the full acceptance set of a decoder that must return  $r$  itself, including the contribution of the admissible denominators  $2r, \dots, \lfloor N/r \rfloor r$  that window bounds do not count. Eq. (5) depends only on  $r$  and  $N$ , not on the base: at matched control dimension the decoder’s tolerance is *identical* across bases, so the entire cross-base difference in decoded success sits in the quantum state—precisely this section’s mechanism claim. We are not aware of this acceptance set having been computed before for a decoder required to return  $r$  itself.

This decomposition has a general reading beyond the present comparison. A decoded success probability conflates two quantities with different physics: the decay of the quantum state, which damage-weighted exposure describes universally, and the error tolerance of the classical post-processing wrapped around the measurement, which is algorithm-specific and can grow with problem size. Benchmarks of “noise resilience” across algorithms or encodings that use decoded success as their metric

are therefore partially measuring decoder redundancy—a classical resource. Reporting end-state fidelity alongside decoded signal, as in Table III, separates the two at no extra simulation cost, and we suggest it as standard practice for noisy-algorithm comparisons.

### VIII. ROBUSTNESS

*Readout error.* A  $d$ -level readout must resolve  $d$  pointer states, the higher ones worst. Charging a readout channel with misread rate of  $|k|$  growing as  $(1+k)$ —a linear reading of the few-percent, higher-is-worse readout errors reported for transmon qutrits [2, 3]—on every control carrier of every base leaves the qudit advantage untouched: the ququint’s lead drifts by less than  $\pm 0.03$  over the full sweep  $\varepsilon = 0\text{--}0.04$  (an  $8\times$  range among nonzero settings). The reason is a structural cancellation: total readout exposure is  $m \times \varepsilon(d+1)/2$ , and at matched precision  $m \approx \log D/\log d$ , giving  $9\varepsilon/8\varepsilon/9\varepsilon$  for  $d = 2/3/5$  at  $D \approx 64$ . Per-level readout degradation with  $d$  is almost exactly cancelled by the reduced carrier count—near-neutral by construction, not luck. Measured hardware sits mid-sweep, not at its edge: transmon qutrit assignment fidelities of 97–99% for  $|0\rangle$  and 92–96% for  $|2\rangle$  [2, 3] correspond to  $\varepsilon \approx 0.01\text{--}0.03$  in the  $(1+k)$  model (Wang’s ten-state single-shot readout at 93.8% [30] sits at its upper end), and at  $\varepsilon = 0.02$  the ladder cells read 0.48/0.72/0.75 for  $d = 2/3/5$  against 0.51/0.73/0.76 at zero—every ordering of Table II intact at the measured operating point. The neutrality claim is scoped to simultaneous  $d$ -level discrimination (the transmon case that motivates the  $(1+k)$  model): ion-qudit readout is sequential shelving with  $d - 1$  detection rounds, so misreads compound down the sequence and readout *time* grows with  $d$ , adding idle decoherence the cancellation above does not include.

*Refocusing.* No transmon runs a long circuit without echo, so the honest question is whom it helps more. Sweeping the dephasing scale from free evolution to perfect echo (the same knob that models the high- $E_J/E_C$  regime [30]), the ququint’s lead grows monotonically in all four conditions tested. The consequential case is linear-in- $d$  gate cost: the ququint *loses* Shor without echo ( $-0.026$ ) and *wins* with it ( $+0.191$ ); QPE moves from exact parity to  $+0.196$ . Mechanistically, dynamical decoupling suppresses dephasing—the part of the ladder channel scaling worst with  $d$  (the max-level law)—and leaves the gentler  $k^{0.7}$  relaxation, in agreement with qudit DD experiments [29] and with the refocused molecular-qudit QFT of Ref. [45]. This is why the central condition is stated “at the device’s operating dephasing level”: refocusing buys roughly one cost model of headroom, and benchmarks of qudit algorithms should be run on refocused devices.

*Noise-inflation threshold.* The channels charge every base the same per-layer strength; if higher- $d$  gates are additionally noisier per layer, the condition acquires a

threshold. Inflating the qudit’s per-layer strength alone,  $s_d = f s_2$ , and locating the crossing  $f^*$  at which the qudit’s floor-corrected signal falls to the qubit’s (demo instance, exact density matrices, both operating points): under native-gate cost  $f^* = 2.0\text{--}2.5$  on the calibrated ladder and 2.6 ( $d=3$ ) to 4.5 ( $d=5$ ) under depolarizing; under the linear ion cost  $f^* = 1.2$  ( $d=3$ , ladder) and 1.6 ( $d=3, 5$ , depolarizing), the ladder ququint cell being already lost at  $f = 1$  (Sec. IV). Read as hardware guidance: a platform keeps the qudit advantage while its measured qudit-to-qubit per-gate noise ratio stays below  $f^*$  after the layer-count multiplier is charged—roughly  $1.6\times$  for today’s trapped-ion pairing and  $2.0\text{--}2.5\times$  for a native-entangler transmon (the calibrated-ladder/uniform pairing; the  $4.5\times$  cell belongs to depolarizing/uniform, a crossed pairing).

*The threshold at measured fidelities.* The only native two-qudit entangling gate characterized across  $d = 2, 3, 5$  on one apparatus—the single-application light-shift gate of Hrmo *et al.* [32], fidelity 99.6(1)/98.7(2)/93.7(3)% at  $d = 2/3/5$ —decides the second condition. Converting a measured infidelity  $\epsilon$  to the implied per-carrier-layer strength must go through the damage identity of the channel being inflated: a strength- $s$  layer deposits entanglement infidelity  $\Delta(d)s$  per carrier, with  $\Delta = 1 - 1/d^2$  for the depolarizing channel but  $\Delta = 0.75/1.46/2.83$  at  $d = 2/3/5$  for the calibrated ladder, whose higher levels already relax and dephase faster (the damage units of Sec. VII). So  $s = \epsilon/[2L\Delta(d)]$  and  $f = s_d/s_2$ ; this conversion is also what keeps the ladder from being charged twice for the same decoherence, since  $f$  then measures the gate’s noise in units of what the ambient ladder itself deposits during the gate’s  $L$  layers. (An earlier version of this analysis applied the depolarizing identity to both channels, overstating the ladder factors by  $1.6\times$  at  $d=3$  and  $3.0\times$  at  $d=5$ .) Under uniform accounting (the gate charged one layer, as its single-application construction warrants)  $f = 1.67 \pm 0.49$  at  $d = 3$  and  $4.17 \pm 1.06$  at  $d = 5$  on the ladder,  $2.74 \pm 0.80$  and  $12.30 \pm 3.13$  under depolarizing; under the ion accounting (the  $(d-1)$  layer charge absorbing part of the growth)  $0.83 \pm 0.24$  and  $1.04 \pm 0.27$  on the ladder,  $1.37 \pm 0.40$  and  $3.08 \pm 0.78$  under depolarizing ( $1\sigma$  propagated from the quoted gate errors). The factor is measured on an entangling gate, so *where* it is charged is a modeling choice, and we bracket it. Global inflation—every carrier-layer at  $f s_2$ , the scope of  $f^*$  above—charges the qudit’s idle and single-qudit layers (two-thirds of the  $d = 5$  exposure) a penalty measured only on its entangler, which the flat measured single-qudit error and  $\sim 100$ -ms shielded idle coherence above argue against; gate-only inflation—the two participants of each entangling gate at  $f s_2$  through that gate’s layers, the rest at  $s_2$ —concedes the qudit an idle noise that does not grow with  $d$  at all. Rerunning the demo grid by exact density matrices at these factors under both scopes (Table V; `results/transmon_rebuild.json`, `results/hrmo_gate_only.json`): *the ququint advantage is lost globally in every channel/cost pairing—the mea-*



sured  $f$  exceeds  $f^*$  by  $1.7\times$  to  $2.7\times$  wherever  $f^*$  is defined, and the ladder/ion cell was lost at  $f = 1$  already—and gate-only in three of four pairings, the exception being the most lenient reading available (ladder, **uniform** cost, entangler-only charge), where it survives centrally by  $+0.04$  and dies at  $+1\sigma$ . *The qutrit now passes under both scopes*: globally it survives three of four pairings (ladder 0.58/0.63 vs the qubit’s 0.51 under **uniform/ion** cost; per-particle depolarizing under **ion** cost 0.38 vs 0.33) and loses only depolarizing/**uniform**, centrally but within  $1\sigma$  (0.32 vs 0.33); gate-only it survives *all four* (0.66/0.60 on the ladder, 0.51/0.44 vs 0.33 under depolarizing). At measured fidelities the second condition therefore fails at  $d = 5$  in seven of eight scope/cost pairings, with the eighth a near-tie, and passes at  $d = 3$  for the trapped-ion gate in seven of eight—the scope question, a measurable property rather than a modeling choice, has moved from deciding the ion qutrit to deciding the transmon one below. One costing convention deserves a sensitivity check: **ion** charges the supplement’s  $2(d-1)$ -pulse construction, while the same reference’s main text quotes  $2d$  Mølmer-Sørensen gates for the controlled-increment. Charging the qudit entangler  $d$  layers instead (the qubit’s controlled-phase remains a single two-pulse gate), both quantities move: the measured  $d = 3$  factor falls to 0.56 (ladder) and 0.91 (depolarizing)—the heavier charge absorbs more of the measured infidelity—while  $f^*$  falls to 0.85/0.69 (ladder,  $d = 3/5$ ) and 1.10/1.33 (depolarizing,  $d = 3/5$ ) (**results/ion\_2d.fstar.json**). The marginal depolarizing/ion qutrit cell survives centrally under either convention ( $0.91 < 1.10$ ), brushing its threshold at  $+1\sigma$  under the harsher one; every  $d = 5$  verdict is unchanged.

*The measured fourth point.* The same reference’s  $d = 4$  gate (97.0(2)%) tests the monotonicity the inflation analysis assumes (**results/hrmo\_d4.json**). The channel-consistent factors interpolate cleanly— $2.61 \pm 0.68$  (**uniform**) and  $0.87 \pm 0.23$  (**ion**) on the ladder, 6.00/2.00 under depolarizing, each between its  $d = 3$  and  $d = 5$  values—and the verdicts interpolate with them: the  $d = 4$  register survives all four ladder readings but only the most lenient depolarizing one (**uniform**, gate-only, 0.36 vs 0.33 centrally), five of eight against the qutrit’s seven and the ququint’s one. The  $d = 3$  outcome is not a boundary artifact, and the analysis’s monotonicity assumption is now measured rather than assumed.

*The transmon side of the same test.* The measured cross-Kerr two-qutrit gates of Ref. [3]—CZ $^\dagger$  process fidelity 97.3(1)%, CZ 95.2(3)%—face the matched calibrated-ladder/**uniform** pairing. Goss *et al.* report no qubit-subspace entangler on the same footing, so the qubit anchor is the operating point  $s_2 = 0.003$  itself, equivalent to a same-class two-qubit gate at 99.55%; the ladder-consistent inflation factors are then  $f = 3.08 \pm 0.11$  (CZ $^\dagger$ ) and  $5.47 \pm 0.34$  (CZ). The CZ fails both scopes even at  $-1\sigma$ . The CZ $^\dagger$  splits them (**results/transmon\_rebuild.json**): globally it is lost even at  $-1\sigma$  ( $f > f^* = 2.05$ ; signal 0.34 against the

qubit’s 0.51), while gate-only it *passes*, robustly at  $\pm 1\sigma$  ( $f = 3.08 < f^* = 3.63$ ; signal 0.55 vs 0.51). Inverted, the transmon targets are a two-qutrit entangler at  $\gtrsim 96.8\%$  (gate-only scope) to  $\gtrsim 98.2\%$  (global) against 97.3% measured: the measured gate sits *inside* the bracket, and the scope question—whether idle decoherence scales with entangling-gate error—now decides the transmon qutrit exactly as it once decided the ion one. On this platform the physics leans to the harsher reading: the charge dispersion of  $|2\rangle$  that drives idle dephasing is the same mechanism that limits the cross-Kerr entangler [2, 3], so idle and gate error should co-scale—the global scope—and the operative target is 98.2%, 0.9 points above the measured gate. Nor is the measured number a ceiling: the same group has since demonstrated the first qutrit error-mitigation protocols—noise tailoring with up to  $3\times$  improvement in effective error on a transmon qutrit processor [46]—so the 0.9-point gap is within reach of demonstrated mitigation, with the caveat that mitigation overhead is itself exposure this paper’s accounting would charge. The verdict is also a function of the assumed anchor: re-anchored to a Willow-class 99.67% two-qubit gate the factor rises to  $f = 4.20$  against  $f_{\text{gate}}^* = 3.57$  and the CZ $^\dagger$  fails both scopes; anchors at 99.34% and 99.0% give  $f = 2.10$  and 1.38 against thresholds that move far less, passing gate-only comfortably and, at 99.0%, the global scope as well. The required two-qutrit fidelity is therefore a schedule against the comparator, not one number: 97.7–98.7% against Willow-class gates down to 92.3–95.7% against a 99.0%-class device (gate-only to global scope; **results/transmon\_rebuild.json** sweeps four anchors with  $f^*$  recomputed at each operating point). Inverting the threshold gives the experimental target: under **ion** accounting the ququint returns to advantage when its native-gate infidelity falls below  $\approx f^* \cdot 4$  ( $0.96/0.75$ )  $\epsilon_2 \approx 0.033$  at  $\epsilon_2 = 0.004$ , i.e. a  $d = 5$  native entangler at  $\gtrsim 96.7\%$  fidelity—against 93.7% measured.

*Wall-clock exposure.* The layer-counted convention above charges the 580-ns cross-Kerr gate one layer—the same as a 30-ns single-qutrit pulse. The opposite convention charges every gate its duration, in units of a reference  $d = 2$  two-qubit gate: (0.2, 1.0) layers for the qubit register’s single/two-qubit gates, (0.3,  $\rho$ ) for the qutrit’s with  $\rho = t_{2q}(3)/t_{2q}(2)$ , every carrier idling through every gate at equal ambient strength per unit time and *no* separate inflation factor—in this convention the entangler’s decoherence is charged as elapsed time, and charging  $f$  on top would double-count from the other side (at the operating point,  $2\rho\Delta(3)s_2 = 0.051$  of ambient damage at  $\rho = 5.8$  already exceeds the whole measured infidelity 0.027). Sweeping  $\rho$  on the demo instance by exact density matrices (**results/transmon\_rebuild.json**), the qutrit’s timed margin over the qubit falls from  $+0.21$  at  $\rho = 1$  through  $+0.04$  at  $\rho = 2$  to  $-0.35$  at  $\rho = 5.8$ , crossing zero at  $\rho^* = 2.3$ : charged its true duration, the Goss gate keeps the advantage only against a two-qubit comparator slower than  $\sim 250$  ns—an IBM-class cross-resonance

TABLE V. The inflation threshold evaluated at the measured native qudit gate of Ref. [32] ( $N=21$  demo instance, exact density matrices; `results/hrmo_reanalysis.json`, ladder rows at the channel-consistent conversion of `results/transmon_rebuild.json`,  $d = 4$  rows from `results/hrmo_d4.json`), under the *global* inflation scope (every qudit carrier-layer at  $fs_2$ ); the gate-only scope, which brackets it from the other side, is reported in the text. “ $f$  measured” is the inflation factor implied by the gate fidelities under each accounting, converted through each channel’s own damage identity ( $\pm 1\sigma$ ); signals are floor-corrected at the central  $f$ , with the qubit anchor at  $s_2 = 0.003$  (ladder) and 0.005 (depolarizing).

| noise  | cost           | $d$ | $f$ measured   | $f^*$ | signal (qubit)     |
|--------|----------------|-----|----------------|-------|--------------------|
| ladder | <b>uniform</b> | 3   | $1.7 \pm 0.5$  | 2.05  | <b>0.58</b> (0.51) |
|        | <b>uniform</b> | 4   | $2.6 \pm 0.7$  | 3.40  | <b>0.60</b> (0.51) |
|        | <b>uniform</b> | 5   | $4.2 \pm 1.1$  | 2.46  | 0.31 (0.51)        |
|        | <b>ion</b>     | 3   | $0.8 \pm 0.2$  | 1.21  | <b>0.63</b> (0.51) |
|        | <b>ion</b>     | 4   | $0.9 \pm 0.2$  | 1.43  | <b>0.67</b> (0.51) |
|        | <b>ion</b>     | 5   | $1.0 \pm 0.3$  | —     | 0.43 (0.51)        |
| depol. | <b>uniform</b> | 3   | $2.7 \pm 0.8$  | 2.65  | 0.32 (0.33)        |
|        | <b>uniform</b> | 4   | $6.0 \pm 1.6$  | 3.64  | 0.12 (0.33)        |
|        | <b>uniform</b> | 5   | $12.3 \pm 3.1$ | 4.50  | 0.04 (0.33)        |
|        | <b>ion</b>     | 3   | $1.4 \pm 0.4$  | 1.56  | <b>0.38</b> (0.33) |
|        | <b>ion</b>     | 4   | $2.0 \pm 0.5$  | 1.53  | 0.22 (0.33)        |
|        | <b>ion</b>     | 5   | $3.1 \pm 0.8$  | 1.62  | 0.12 (0.33)        |

gate ( $\rho \approx 1.9$ ), not a 100-ns CZ ( $\rho = 5.8$ ), and not a Google-class 30-ns gate ( $\rho \approx 19$ , margin  $-0.71$ ). The layer-counted and timed conventions bracket the physical schedule from opposite sides; that the transmon qutrit passes one and fails the other at today’s gate times is the honest state of the platform, and the  $d \geq 5$  verdicts only worsen here, no transmon entangler beyond  $d = 3$  having been demonstrated at all [3].

*Leakage.* The ladder channel is trace-preserving on the computational manifold; a real cross-Kerr gate is not, and part of the measured  $\epsilon_3$  is leakage out of  $\{|0\rangle, |1\rangle, |2\rangle\}$ . If a fraction  $\lambda$  leaks, the in-manifold inflation drops to  $(1 - \lambda)f$  while the leaked population, scored at the decoder floor, multiplies the signal by  $\sim(1 - \lambda\epsilon_3)^{18}$  over the demo circuit’s 18 two-qutrit interactions. At the  $\lambda = \frac{1}{2}$  endpoint this trades  $f = 3.08 \rightarrow 1.54$  against a 0.78 survival factor: the gate-only pass thins from 0.55 to 0.53 against the qubit’s 0.51 and the global loss stands (0.47). Leakage does not flip either verdict at the demo depth, but it is a qudit-only charge with no qubit counterpart, it compounds with depth ahead of any in-manifold error, and its measured fraction on this gate is the single most useful number a transmon follow-up could report.

*Ladder-exponent sensitivity.* The calibrated ladder fixes relaxation at  $\Gamma_k \propto k^{0.7}$  and dephasing at  $\max(j, k)^{1.1}$ . Both exponents are constrained mainly by  $d = 3$  pair data (Blok’s 1 : 2.0 : 2.3, which the channel realizes as 1 : 2.14 : 2.14), and are then applied at  $d = 5$

TABLE VI. Sensitivity of the ladder verdicts to the per-level exponents: floor-corrected signal on the demo instance at  $s = 0.003$ , exact density matrices. “paper” is the calibration used throughout; the remaining rows steepen dephasing toward the range Peterer *et al.* require, the last also moving relaxation to their  $k^{1.0}$ . Boldface marks the best base in each row.

| cost           | exponents ( $\Gamma_k / \Gamma_\phi$ ) | $d=2$ | $d=3$        | $d=5$        |
|----------------|----------------------------------------|-------|--------------|--------------|
| <b>uniform</b> | 0.7 / 1.1 (paper)                      | 0.506 | 0.725        | <b>0.760</b> |
|                | 0.7 / 1.6                              | 0.506 | <b>0.697</b> | 0.666        |
|                | 0.7 / 2.0                              | 0.506 | <b>0.667</b> | 0.556        |
|                | 0.7 / 2.6                              | 0.506 | <b>0.609</b> | 0.342        |
|                | 1.0 / 2.6                              | 0.506 | <b>0.604</b> | 0.328        |
| <b>ion</b>     | 0.7 / 1.1 (paper)                      | 0.506 | <b>0.568</b> | 0.442        |
|                | 0.7 / 1.6                              | 0.506 | <b>0.528</b> | 0.299        |
|                | 0.7 / 2.0                              |       | <b>0.506</b> | 0.488        |
|                | 0.7 / 2.6                              |       | <b>0.506</b> | 0.415        |
|                | 1.0 / 2.6                              |       | <b>0.506</b> | 0.410        |

and 7. A provenance caveat sharpens the stakes: if those pair ratios were extracted from Ramsey decay in a quasi-static-dominated regime—where decay is Gaussian and  $1/T_2^*$  scales with the *first* power of the level sensitivity—then the Lindblad rates realize their *squares*, 1 : 4.0 : 5.3, a steepening the sweep below prices at dephasing exponent  $\approx 2$ . Peterer *et al.* [28], who measured the sequential ladder to the fourth level, disagree in both places: their  $T_1$  series gives  $\Gamma_{21}/\Gamma_{10} = 2.05$  (i.e.  $k^{1.0}$ , not  $k^{0.7}$ ), and their  $T_2$  series gives pair rates 1 : 2.25 : 6.0 :  $>36$  against 1 : 2.14 : 3.35 : 4.59 under the max-level law. No power law fits that escalation—exponent 1.6 matches the  $2 \leftrightarrow 3$  pair, 2.6 matches  $3 \leftrightarrow 4$ , neither matches both—so rather than refit we sweep the exponent across the range Peterer admits and ask what survives (Table VI). The  $d = 2$  column is a control: the channel is normalized on the  $0 \leftrightarrow 1$  subspace, so every  $d = 2$  cell must be invariant, and is.

Two things survive and one does not. Under native-gate cost the *qutrit* advantage is robust: it holds at every exponent tested, degrading only from 0.725 to 0.604 across the full range while the qubit sits at 0.506. The *ququint* advantage is not robust: it is already gone at the first steepening (0.666 against the qutrit’s 0.697 at exponent 1.6) and falls below the qubit by exponent 2.6. Under linear **ion** cost even the qutrit advantage fails once dephasing reaches exponent 2.0. A trajectory run extending the same sweep to  $d = 7$  (1000 per point; its ten independent  $d = 2$  draws scatter about the exact 0.5063 with mean  $z = -0.7$ , the worst at  $-2.6\sigma$ ) shows the pattern continuing:  $d = 7$  falls from 0.682 to 0.075 under **uniform** cost across the range, statistically tied with the qubit at exponent 1.6 ( $0.504 \pm 0.024$  against the exact 0.506) and clearly below from 2.0 on ( $0.345 \pm 0.025$ ,  $>4\sigma$ ). The dimension at which qudits help is therefore bounded by the very exponent the data constrain least: on the calibrated ladder as published,  $d = 5$  leads; on a Peterer-consistent reading, only  $d = 3$  survives, and only

with a native gate. We report the paper’s grid at the published calibration and flag every  $d \geq 5$  ladder result as conditional on it.

*Dephasing weight.* A second unexamined constant sits beside the exponents: the channel fixes the pure-dephasing weight so that  $T_2 = T_1$  on the  $0 \leftrightarrow 1$  subspace, while measured transmons span roughly  $T_2 \approx 0.3T_1$  to  $2T_1$ . Sweeping the dephasing-to-relaxation weight over 0.2–5—i.e.  $T_2/T_1$  from 1.67 down to 0.33—and re-scoring the ladder rows of Table II at their  $s = 0.005$  operating point (exact density matrices; `results/dephase_ratio_sweep.json`) moves the verdicts only at the dephasing-dominated end. At fixed strength every qudit-vs-qubit verdict survives the full span; the only change is internal to the qudits (at  $T_2 = T_1/3$  the **uniform** optimum moves from  $d = 5$  to  $d = 3$ ) and the **pavlidis** qutrit margin thins to  $+0.004$ . The sharper reading rescales  $s$  so the *qubit’s* per-carrier-layer damage stays at its calibrated value, isolating the  $d$ -dependent structure of dephasing from the overall noise level. There the native-gate qutrit survives everywhere (0.55 against the qubit’s 0.37 at  $T_2 = T_1/3$ ), but the **ion-cost** qutrit win inverts at  $T_2 = T_1/3$  (0.34 vs 0.37), the thin **pavlidis** win is already a tie at  $T_2 = 0.67T_1$ , and the ququint’s **uniform** lead over the qutrit inverts at  $T_2 = T_1/3$ . The pattern repeats the exponent sweep’s: dephasing is the component of the ladder that scales worst with  $d$ , so every advantage except the native-gate qutrit’s is additionally conditional on dephasing not dominating relaxation—the upper half of the measured transmon  $T_2$  range.

*Quasi-static and correlated dephasing.* The ladder’s stated microscopic mechanism—charge dispersion—is  $1/f$  and quasi-static on gate timescales, and transmons sharing a feedline or TLS bath acquire spatially correlated dephasing, so the substitution that withdrew the Zeeman reversal (Table VIII) is mandatory here too. The calibrated dephasing being realized by diagonal jumps, its Euclidean level vectors double as quasi-static sensitivities: a static offset drawn once per shot—per carrier, or one for the register—reproduces the calibrated pair structure exactly, with damage growing as depth<sup>2</sup>, matched to the Markovian  $0 \leftrightarrow 1$  damage on the qubit’s **uniform** circuit and then held fixed across bases; relaxation stays local and Markovian throughout (`results/ladder_quasistatic.json`; exact DM for the Markovian rows, 1000 trajectories otherwise). All four temporal $\times$ spatial combinations preserve every verdict of the  $s = 0.003$  demo cells, and both referee-predicted signs are confirmed. Quasi-static dephasing lifts every base but the shallow qudit schedules most—the **uniform** ququint from 0.760 to 0.863(10), its margin over the qubit widening from 0.254 to 0.301—so the Markovian stand-in is *conservative* for the qudits, as the ion analysis already implied. Common-mode correlation helps the wide qubit register most ( $+0.06$  against the ququint’s  $+0.006$ ), thinning the **ion** qutrit margin from 0.062 to 0.046 without flipping it. The paper’s Markovian, uncorrelated con-

vention is therefore mildly pessimistic for qudits on the temporal axis and mildly optimistic on the spatial one; neither axis, nor their combination, moves a verdict.

*Thermal excitation and leakage.* The ladder as calibrated is downward-only and truncated at  $d - 1$ : no thermal excitation (transmons run at  $\bar{n} \sim 0.01$ –0.05) and an artificially confined top level. Both omissions have a qudit-favoring *per-carrier* sign, upward matrix elements growing as  $\sqrt{k}$ . Extending every carrier to  $d + 1$  levels—detailed-balance up-jumps at rate  $\bar{n}\Gamma$ , the top rung being leakage out of the encoded manifold, gates acting as identity on the leak level, any leaked outcome decoded as a failure—and sweeping  $\bar{n}$  to 0.4 at the  $s = 0.003$  **uniform-cost** operating point (`results/ladder_thermal.json`; 1000 trajectories per point, the  $\bar{n} = 0$  column reproducing the exact leak-free values within errors) shows the register-level sign is the opposite: the *qubit* loses most. Its  $11 \times 57$  carrier-layers of thermal exposure against the ququint’s  $5 \times 15$  outweigh the per-carrier growth: at the physical  $\bar{n} = 0.05$  the qubit falls  $0.52 \rightarrow 0.47$  while both qudit columns move within their error bars, and at  $\bar{n} = 0.4$ —an order of magnitude beyond hardware—the qubit is destroyed (0.04) while the ququint still holds 0.69. Neither the  $d = 3$  nor the  $d = 5$  advantage crosses anywhere in the swept range; the gaps *widen* monotonically,  $+0.21 \rightarrow +0.55$  and  $+0.24 \rightarrow +0.65$ . Confining the top level was a bias against the wide qubit register, not for the qudits.

*Concurrent schedules.* Every result here is an exposure budget, and the budget charges every carrier an idle channel through every gate. That serial convention is not neutral: concurrent execution shortens the schedule most for the register with the most carriers, which is  $d = 2$ , so the assumption runs *against* the conclusion it supports and deserves a measurement rather than a disclaimer. Packing the gate list by as-soon-as-possible list scheduling on carrier availability—exact, since gates on disjoint carriers commute—compresses the demo circuits by 26.3%/23.1%/20.0% at  $d = 2/3/5$  under **uniform** cost, 26.3%/15.9%/7.1% under **ion**, and 26.3%/14.9%/4.8% under **pavlidis**, confirming the predicted direction: the qubit gains most, and under the surcharged costs the ququint gains almost nothing. Re-running the six cells of Table II under the parallel schedule leaves five of the six qudit-vs-qubit verdicts unchanged while every margin erodes:  $+0.452 \rightarrow +0.385$  and  $+0.172 \rightarrow +0.113$  (depolarizing, **uniform/ion**),  $+0.349 \rightarrow +0.268$  and  $+0.092 \rightarrow +0.023$  (ladder), and  $+0.130 \rightarrow +0.076$  (depolarizing/**pavlidis**). The one flip is the ladder/**pavlidis** cell, whose thin serial qutrit win ( $+0.051$ ) becomes a thin qubit win ( $-0.020$ )—the same cell the single-qudit charge below eliminates, by the same logic: its serial margin lives on uncharged qudit layers. Two further readings follow. The serial convention inflates the qudit margin by up to a factor of four but does not create it where the win is real; and the ladder/**ion** cell, at  $+0.023$  under concurrency, is thin enough that it should be read as a tie rather than a win. The depolarizing/**ion** cell independently repro-

duces the near-tie found in the strength sweep above: under concurrency its best qudit switches from  $d = 5$  to  $d = 3$ .

*Single-qudit gate cost.* Both **uniform** and **ion** charge a single-qudit gate one layer at every  $d$ . Per laser pulse that matches Ref. [1]; per *operation* it does not, since the same device’s randomized benchmarking gives a per-Clifford error rising  $5\times$  from  $d = 3$  to  $d = 5$  on  $\sim 3\times$  the pulses. The omission matters more than it looks, because the single-qudit share of the schedule *grows* with  $d$ : the order-finding circuit spends  $2m$  of its layers on single-qudit Fourier rotations, which is 21%/31%/40% of the serial schedule at  $d = 2/3/5$ . A  $d$ -dependent charge therefore hits the qudit twice. Charging single-qudit gates  $\mu_1(d) = (d/2)^\alpha$  layers, normalized to 1 at  $d = 2$  like every other multiplier here, and sweeping  $\alpha$ , the measurement pins the  $d=3 : d=5$  ratio to  $\alpha = 2.15$  (pulse count) or  $\alpha = 3.15$  (per-Clifford error); the  $d = 2$  end is an extrapolation, which is why this is reported as a threshold rather than a table row. The charge is not an ion quirk: an independent Rydberg-superatom qudit encoding derives its generalized Fourier gate at  $\sim (d/2)^2$  laser pulses— $\alpha = 2$  by construction, before any entangling gate exists on that platform [7]. Four of the six cells survive the steeper calibration: depolarizing under **uniform** ( $\alpha^*$  beyond the swept range), **ion** ( $\alpha^* = 3.55$ ), and—by a hair—**pavlidis** ( $\alpha^* = 3.17$ , margin  $+0.130 \rightarrow +0.056$  at the pulse-count charge and  $+0.001$  at the per-Clifford charge), and the ladder under **uniform** ( $\alpha^* = 3.84$ ). Two do not, and both are ladder cells. The ladder/**ion** cell has  $\alpha^* = 2.14$ —it fails at *exactly* the measured pulse-count charge, margin  $+0.092 \rightarrow -0.001$ . And the ladder/**pavlidis** cell, whose serial qutrit win is new to the corrected decomposition charge, has  $\alpha^* = 1.44$  and does not survive any measured charge ( $+0.051 \rightarrow -0.036$  at  $\alpha = 2.15$ )—the same cell concurrency flips, for the same reason. The depolarizing/**pavlidis** survival is the analysis’s most delicate verdict and we read it as a tie at the steep end rather than a win. Across the whole swept range the best qudit is  $d = 3$  in every cell from  $\alpha \geq 1.5$ —by  $\alpha = 3.15$  the ququint has fallen below the qubit even under native-gate cost—which is the same conclusion the ladder-exponent, dephasing-weight and measured-fidelity analyses reach independently.

*Zeeman-structured dephasing.* The per-particle convention flattens pair structure the ion encoding really has, so we also test its worst case: magnetic-field dephasing carrying the collective- $B$  sensitivity structure of the  $^{40}\text{Ca}^+$  level indexing of Ref. [1], applied independently per carrier—the uncorrelated worst case—realized exactly by a single diagonal jump  $\propto \text{diag}(g_j m_j)$  ( $g_S = 2$ ,  $g_D = 6/5$ ; the Markovian generator stands in for the unrefocused effect of what is, in the laboratory, quasi-static noise), normalized so the optical-qubit pair  $0 \leftrightarrow 1$  matches the qubit channel. Pair rates then span  $1\text{--}25\times$  that reference at  $d = 3$  and  $1\text{--}49\times$  at  $d = 5$ . The encoding is no strawman: over all  $\binom{8}{d}$  level subsets of the  $S_{1/2}/D_{5/2}$  manifold, the chosen levels exactly minimize the worst

TABLE VII. Zeeman-structured dephasing on the demo instance under a *Markovian* generator: floor-corrected signal at  $s = 0.003$ , **uniform** cost. The Zeeman jump is the *only* channel in this table (a structured-dephasing stress test, not composed with a base noise model; Table VIII instead composes it with the depolarizing operating point, which is why the  $d = 2$  columns of the two tables differ). Both variants reverse the qudit ordering of Table II, and the reversal persists under **ion** cost—but it does not survive replacing the generator with quasi-static field noise (Table VIII).

| dephasing structure | $d=2$       | $d=3$ | $d=5$ |
|---------------------|-------------|-------|-------|
| per-carrier Zeeman  | <b>0.72</b> | 0.24  | 0.23  |
| common-mode Zeeman  | <b>0.87</b> | 0.45  | 0.34  |

pair rate at  $d = 5$  and  $d = 7$ ; only  $d = 3$  admits a gentler choice ( $9\times$ , via  $\{S_{-1/2}, D_{-3/2}, D_{-1/2}\}$ ).

The verdict reverses outright (Table VII): the qubit wins at every strength under both cost models. Nor is the missing correlation structure doing the work: repeating the study with genuinely common-mode dephasing—one global jump coupling to the total Zeeman shift, exact as an elementwise mask on the density matrix—preserves the reversal; its decoherence-free pairs soften every base, the qubit most. Unmitigated *Markovian* Zeeman-structured dephasing on a sensitivity-ordered encoding is therefore a regime with no qudit advantage at all—the sharpest failure mode we found, and one the condition’s “operating dephasing level” clause must be read to include. That qualifier carries the weight, however: the Lindblad generator is a stand-in for noise that is quasi-static, and replacing it with the real thing changes the answer in the qudit’s favor.

*Refocusing is  $d$ -dependent on this encoding.* A static field offset is removed by permuting levels: a sequence  $\pi_1, \dots, \pi_L$ , each held for an equal share of the idle window and undone at the end, leaves level  $j$  with mean coefficient  $\tilde{c}_j = L^{-1} \sum_l c_{\pi_l(j)}$ , and refocuses exactly iff  $\tilde{c}$  is constant in  $j$ . For the  $^{40}\text{Ca}^+$  coefficients this has a sharp answer. At  $d = 2$  the level swap gives  $\tilde{c}$  constant: one  $\pi$ -pulse cancels the qubit’s dephasing exactly, as the standard argument says. At  $d \geq 3$  *no* two-interval echo refocuses, for any permutation; the best leaves a residual pair splitting of  $1.5\times$  ( $d = 3$ ) or  $1.0\times$  ( $d = 5, d = 7$ ) the *unrefocused qubit* splitting. Exact refocusing needs  $L = d$  intervals—the cyclic shift—and  $d$  is minimal over all permutation sequences, not merely over powers of one pulse (exhaustive search,  $d = 3$  and  $d = 5$ ). The mitigation that dismisses this regime is therefore free for the qubit and costs  $d$  pulses for the qudit, which is the opposite of the  $d$ -independence the constant-overhead passive-decoupling construction [47] would supply, and the reason that construction is the right thing to reach for here.

*The Markovian stand-in was not conservative.* Simulating the field as it is—one offset per shot, drawn from a Gaussian and held fixed, integrated over that Gaussian by quadrature rather than sampled—moves every



TABLE VIII. Quasi-static versus Markovian field noise, with and without refocusing: floor-corrected signal on the demo instance at the depolarizing operating point  $s = 0.005$ , with a common-mode Zeeman component in both cases. The field width  $\sigma$  is fixed once by matching the Markovian  $0 \leftrightarrow 1$  damage over the  $d = 2$  circuit and then held fixed across bases, so the rows differ only in the time statistics of the same field. “echo” is the best two-interval sequence, “ $d$ -pulse” the exact  $d$ -interval one, and “charged” pays its pulses as exposure—one extra single-qudit layer per pulse *per gate* (the naive  $d$ -interval sequence over an  $\sim 50$ -gate schedule roughly doubles the wide registers’ exposure, which is why the  $d = 2$  margin drops so sharply).

| field        | mitigation | uniform |              |              | ion          |              |              |
|--------------|------------|---------|--------------|--------------|--------------|--------------|--------------|
|              |            | $d=2$   | $d=3$        | $d=5$        | $d=2$        | $d=3$        | $d=5$        |
| Markovian    | none       | 0.275   | <b>0.299</b> | 0.264        | <b>0.275</b> | 0.160        | 0.052        |
| quasi-static | none       | 0.296   | 0.539        | <b>0.605</b> | 0.296        | <b>0.304</b> | 0.181        |
| quasi-static | echo       | 0.331   | 0.622        | <b>0.778</b> | 0.331        | 0.402        | <b>0.479</b> |
| quasi-static | $d$ -pulse | 0.331   | 0.667        | <b>0.782</b> | 0.331        | 0.497        | <b>0.502</b> |
| quasi-static | charged    | 0.173   | <b>0.387</b> | 0.361        | 0.173        | <b>0.283</b> | 0.232        |

unmitigated cell toward the qudit (Table VIII). Under native-gate cost the verdict goes from near-parity (0.275 vs 0.299) to a clear ququint win (0.296 vs 0.605); under linear cost it goes from an outright qubit win (0.275 vs 0.160) to a tie (0.296 vs 0.304). The mechanism is the exponent, not the scale: quasi-static damage accumulates as the *square* of circuit depth, and once  $\sigma$  is pinned on the qubit’s 57-layer schedule the shallower qudit schedules (26 and 15 layers under **uniform**) are rewarded superlinearly. The reversal reported above is thus an artifact of the substitution the manuscript flagged, and we withdraw it as a claim about laboratory field noise while retaining it as what it is—a Markovian worst case. Two internal checks support the arithmetic: the  $d$ -pulse rows reproduce the pure-depolarizing signals of Table II exactly, as they must once the Zeeman component is cancelled, and the quadrature is converged to  $2 \times 10^{-14}$  under halving of its node spacing.

*Charging the pulses returns the advantage to  $d = 3$ .* Idealizing refocusing pulses as free—the most generous reading of the passive constructions—the ququint leads in every mitigated cell. Charging each pulse one layer of exposure, which is what the naive  $d$ -interval sequence costs, reverses that:  $d = 3$  overtakes  $d = 5$  under both cost models (0.387 vs 0.361 and 0.283 vs 0.232), while both stay above the qubit’s 0.173. This is the fourth independent route in this paper to the same place—the qutrit, not the ququint, is what survives realistic accounting.

Measured devices are engineered away from this regime: shielding makes coherence times of order 100 ms achievable across all transitions [1]. The device’s randomized benchmarking—per-pulse error rising only  $1.6 \times$  from  $d = 3$  to  $d = 5$  ( $2.0 \times 10^{-4}$  to  $3.2 \times 10^{-4}$ ) where the raw sensitivity ratio is 25–49—is consistent with a near-flat residual, though a few-microsecond pulse can-

not resolve slow dephasing; the direct evidence the convention needs is  $d$ -resolved Ramsey or echo coherence across the encoding’s transitions, and pending that, the per-particle convention is a statement about mitigated hardware whose price of failure the following sweep fixes. The mechanism mirrors the refocusing result above—structured, slow dephasing is exactly the part echoes remove—and the price is quantitative: composing the Zeeman component with the depolarizing operating point ( $s = 0.005$ ; the two channels commute, so the product is exact) and sweeping the suppression factor  $\varepsilon$  applied to the unmitigated Zeeman reference ( $s_Z = 0.003$  per carrier-layer), the qudit ordering returns at  $\varepsilon^* = 0.58$ – $0.79$  under native-gate cost but only at  $\varepsilon^* = 0.09$ – $0.15$  under the  $2(d-1)$  Mølmer-Sørensen cost. In hardware units the bar is a  $0 \leftrightarrow 1$ -pair coherence of  $\gtrsim 400$ – $600$  layer times for native-gate cost and  $\gtrsim 2200$ – $3800$  for MS cost: at  $\sim 100 \mu\text{s}$  layers, a 100 ms shielded coherence meets the native-gate bar with a factor-two margin but falls short of the MS-cost bar by  $2$ – $4 \times$ . On linear-cost hardware the Zeeman component must be suppressed below the plain shielding level—deeper echo sequences, passive decoupling of the full manifold [47], or magnetically insensitive encodings—before the qudit ordering returns; this  $\varepsilon^* = 0.09$ – $0.15$  bar is the actionable specification this subsection contributes to the experimental proposal of Sec. X.

*Composite dimension.* Nothing in the circuits or channels uses primality. A  $d = 4$  point on the demo instance lands in the qudit band (0.83/0.75 against 0.73/0.67 at  $d = 3$  and 0.76/0.78 at  $d = 5$ ), but  $d = 4$  is a prime power— $\text{GF}(4)$  exists, so the fault-tolerant machinery does not break there—and its  $D = 64$  register shares the qubit’s residual misalignment bit for bit, so it cannot settle the question alone. The decisive control is  $d = 6$ —composite, non-prime-power, no field structure—run on the alignment-neutral instance ( $N = 29$ ,  $r = 7$ ; residual misalignment 0.2857, identical for every base  $d = 2$ – $6$ ; 1000 trajectories per point): the family reads 0.44/0.62/0.73/0.71/0.77 ( $d = 2/3/4/5/6$ , calibrated ladder) and 0.28/0.53/0.72/0.73/0.83 (depolarizing), both composites inside the qudit band and  $d = 6$  at its top under depolarizing. The bare-circuit dynamics carries no trace of primality; the prime restriction elsewhere in this paper is inherited from the fault-tolerance and QFT-arithmetic motivations [8, 10, 11].

*Independent validation.* Beyond internal checks (Sec. XI), we reproduce the three central equations of Ref. [34]—qudit and multi-qubit process infidelities and the critical curve  $(d^2 - 1)/(3 \log_2 d)$ —from our superoperator code with no analytics of our own, to a worst relative error of  $4.1 \times 10^{-4}$  over  $d = 2$ – $64$ , with the residual identified as their first-order truncation (it tracks the infidelity itself).

TABLE IX. Hardware runs vs compiled-circuit predictions. “predicted” is the exact-DM band with the vendor-reported two-qubit infidelity used directly as the per-layer strength ( $s = 0.007$  for Forte-1), spanning the two exposure accountings; the paper’s own gate-to-strength conversion  $s = \epsilon/1.5$  shifts each band upward (see text); Garnet bands are computed on the *routed* circuits recovered from the task metadata (47 and 104 native CZ gates after server-side SWAP routing; `results/garnet_routed.json`), at the same  $s = 0.007$  spanning the same two accountings. Forte-1 main runs use IonQ debiasing over 5,000 shots; success is scored in the digit-reversed reading (see text).  $|w\rangle$  is the work-qubit  $|1\rangle$  population.

| device  | circuit            | predicted | measured          | $ w\rangle$ |
|---------|--------------------|-----------|-------------------|-------------|
| Forte-1 | $m=5$ (15 gates)   | 0.60–0.70 | $0.617 \pm 0.007$ | 0.99        |
| Forte-1 | $m=7$ (28 gates)   | 0.42–0.54 | $0.011 \pm 0.001$ | 0.99        |
| Forte-1 | $m=7$ AQFT (25)    | 0.44–0.57 | $0.066 \pm 0.004$ | 0.99        |
| Garnet  | $m=5$ (47 routed)  | 0.18–0.42 | $0.080 \pm 0.004$ | 0.81        |
| Garnet  | $m=7$ (104 routed) | 0.04–0.15 | $0.032 \pm 0.003$ | 0.66        |

## IX. HARDWARE ANCHOR

The simulation stack makes falsifiable predictions, and the qubit rows of Table X compile at face-value gate counts on cloud hardware: with the work register reduced to one qubit prepared in the eigenstate, each controlled- $U^{2^i}$  is a single controlled-phase, and the no-swap inverse QFT adds  $m(m-1)/2$  more—15 entangling gates at  $m=5$  (6 qubits), 28 at  $m=7$  (8 qubits). We ran these circuits on two commercial devices (Table IX): the trapped-ion IonQ Forte-1 (all-to-all connectivity, vendor-reported two-qubit  $ZZ$  fidelity 0.993 at the August 2026 calibration current when the tasks ran—a figure that drifts between calibrations) and the 20-qubit square-lattice superconducting IQM Garnet. Predictions were recomputed for the compiled circuits by exact density-matrix evolution under the paper’s per-particle depolarizing convention, bracketed by two exposure accountings (every gate a full noise layer, or single-qubit gates charged their measured 130/970 duration ratio).

Three findings. *First, the shallow ion circuit fixes the channel’s scale to within the conversion ambiguity:* the measured  $0.617 \pm 0.007$  sits inside the 0.60–0.70 band when the vendor infidelity is read directly as the per-layer strength ( $s = 0.007$ ), and  $\approx 0.07$  below the 0.69–0.76 band that follows from the paper’s own gate-to-strength conversion  $s = \epsilon/1.5 = 0.0047$  (the convention Sec. VIII applies to the Hrmo gate)—the two readings differ by exactly the factor  $2L(1 - 1/d^2)$ , and one scalar cannot decide between them. Debiasing is an error-mitigation ensemble average over qubit mappings and is not doing the work here: the raw 1,000-shot pilot scored  $0.608 \pm 0.015$ , within 0.009 of the debiased value. Because the success-vs-strength map is steep, this single number is consistent with an effective per-gate depolarizing strength of 0.005–0.010 across the two exposure conventions, a range that brackets the vendor’s measured 0.7% two-qubit infidelity

but sits above the 0.0047 our own conversion assigns it. We read this as a single-point consistency check of the channel’s *scale* at shallow depth—one scalar cannot validate the channel’s structure or pin the conversion, and SPAM, single-qubit error, and idle decoherence are all absorbed into the one fitted parameter, plausibly accounting for the excess over the converted prediction—while the two deeper circuits below bound the depth range over which the check holds. *Second, the deep ion circuit fails coherently, not by decoherence.* At  $m=7$  the interference peak is destroyed (0.011, below the random floor of 0.031) while the work qubit remains at 0.99; no relabeling of the outcome bits recovers the peak (the best over 10,080 reinterpretations—every ordering of the control bits in both polarities—reaches only 0.15, not significant over so large a hypothesis set), and a 500-shot raw  $m=4$  probe caught the mechanism directly, unmitigated—a nearly pure output state (71% of shots on one outcome) whose phase is wrong by one least-significant digit. Debiasing does not remove the failure the raw probe establishes, and an approximate-QFT variant that drops the three smallest rotation angles recovers only a factor  $\sim 6$  (0.066 vs a 0.44–0.57 band): the deep no-swap QFT fails wholesale on this device, consistent with accumulated coherent angle systematics rather than the incoherent noise our channels describe. (A base-2 datum adjacent to the truncation-robustness conjecture for qudits [16]—which it cannot test directly: truncation buys an order of magnitude but cannot repair a coherent failure.) *Third, the superconducting lattice fails mostly—but not only—by routed decoherence.* Recovering the routed circuits from the task metadata turns what an earlier version of this section could only infer from work-qubit decay into a test (`results/garnet_routed.json`): SWAP-routing the all-to-all kickback pattern through the fixed-connectivity chip inflates 15 and 28 entangling gates to 47 and 104 native CZs, and exact-DM depolarizing over those routed circuits (parse validated by reproducing the analytic noiseless successes to  $10^{-6}$ ) predicts the decayed work-qubit populations at  $s \approx 0.004$  per layer. But conditioned on that work-qubit-pinned scale, plain decoherence predicts control success 0.31 ( $m=5$ ) and 0.09 ( $m=7$ ) against the measured 0.080 and 0.032: a 3–4 $\times$  coherent excess on the control register at both depths, the same signature the ion device shows at depth. Routing overhead sets the scale of the failure; a coherent residual finishes it.

The three-way outcome maps onto the paper’s framework. The per-particle depolarizing convention quantitatively describes the platform it was calibrated to—provided the circuit is shallow enough that coherent systematics stay subdominant, the regime our Limitations already fence off. The superconducting failure is compilation overhead deciding the outcome before noise structure enters, the hardware face of the `pavlidis` lesson. And the deep-circuit ion failure is a caution for any decoded-success benchmark on NISQ hardware: past a depth threshold, what is measured is coherent calibration

error, not the decoherence such benchmarks nominally probe. One practical note for reproduction: IonQ returns the control register digit-reversed relative to the Braket local-simulator convention (the  $m = 5$  peak sits exactly on the bit-reversed ideal outcome); all scores above use that pre-registered reading.

## X. DISCUSSION

*Which platform first.* Both physically matched cost/noise pairings favor ququints under their mitigated operating conventions, but the two conditions now pull the platforms apart. Structurally, the native-entangler route is the robust one: it tolerates a Zeeman component up to  $\varepsilon \approx 0.6$ – $0.8$  of the unmitigated reference where the linear-MS-cost route tolerates only  $0.09$ – $0.15$  (Sec. VIII). At measured fidelities the ion qutrit is the stronger case: its gate passes globally in three of four pairings and gate-only in all four, while the transmon entangler passes only the gate-only scope ( $f = 3.08 \pm 0.11$  against  $f_{\text{gate}}^* = 3.63$  but  $f_{\text{glob}}^* = 2.05$ ; Sec. VIII)—a scope the platform’s shared charge-dispersion mechanism argues against—and only under layer-counted exposure, its advantage surviving wall-clock charging only against two-qubit comparators slower than  $\sim 250$  ns. The structurally robust route thus waits on either a  $\gtrsim 98.2\%$  two-qutrit entangler at its own anchor or a demonstration that transmon idle noise does not track entangler error. The ion route also offers the larger headline gains on eigenstate QPE, conditional on collective- $B$  suppression beyond plain shielding—a requirement that belongs to the experimental specification below. The cheapest decisive experiment we can propose is eigenstate QPE at  $m = 2$ – $3$  on a Ringbauer-class ion processor [1], where multi-qudit entangling algorithms have been demonstrated at  $d = 3$  and  $d = 5$  (a qutrit-quint lattice-gauge simulation [48]; earlier  $d = 3$  circuit work in Refs. [49, 50]); single-qudit demonstrations reach  $d = 8$  on a trapped ion [50] and, on photonic platforms,  $d = 32$  Shor and qudit QPE [51, 52]. Table X supplies the predicted success probabilities at common per-base strengths, computed by exact density-matrix evolution with no sampling error. Read honestly against measured gates, however, the table must be entered *per base*, and the measured-fidelity analysis of Sec. VIII then predicts the  $d = 5$  experiment *fails* on today’s hardware: at the strengths the Hrmo gate set implies ( $s_5 = 0.0082$  under the ion accounting,  $s_2 = 0.0027$  from the same apparatus’s  $d = 2$  gate), the qubit register wins both pairs— $0.657$  vs  $0.599$  at 4 bits and  $0.533$  vs  $0.427$  at 5 bits (`results/qpe_measured_strengths.json`)—and the ququint recovers the 4-bit pair only once its native-gate fidelity clears  $\approx 96.7\%$  (Sec. VIII), against  $93.7\%$  measured. The proposal therefore splits into a near-term and a target-setting branch. Near term, the decisive test is the *qutrit*: at measured fidelities it survives *both* inflation scopes in three of four channel/cost pairings, including the matched trapped-ion one (ion cost,

per-particle noise,  $f = 1.37 < f^* = 1.56$  globally, comfortable gate-only); the margins are small enough that either outcome is informative, and the circuits are shallower than anything in Table X. Charging each base the strength its own measured gate implies ( $s_3 = 0.0037$ ,  $s_2 = 0.0027$ ), eigenstate QPE at  $d = 3$ ,  $m = 4$  (ten entangling gates,  $D = 81$ ) is predicted to succeed at  $0.612$  against  $0.533$  for the matched  $d = 2$ ,  $m = 7$  register (twenty-eight gates,  $D = 128$ ); the qutrit’s raw-success margin survives the alternative  $d = 2$ ,  $m = 6$  pairing ( $0.612$  vs  $0.505$ ), though its floor-corrected signal does not ( $0.600$  vs  $0.603$ ), the coarser  $D = 64$  grid depressing that register’s noiseless ceiling to  $0.816$ . Raw success is the directly measured quantity and the one we would score. The shallower  $m = 3$  pair is a dead heat by construction ( $0.655$  vs  $0.657$  raw, signal  $0.734$  vs  $0.695$ ) and serves as a systematics check rather than a test (`results/qpe_d3_measured.json`).

*The error budget in hardware units.* The channel behind these predictions charges gates and serial idle time; four further budget items an experimental run must carry are estimable from the cited devices, and we state them with their signs. (i) *Single-qudit rotations*: the  $\sim 2m$  Fourier rotations per circuit are charged one layer each at the per-pulse error; at the measured per-Clifford error ( $2 \times 10^{-3}$  at  $d=3$  [1]) the excess is priced by the single-qudit-cost sweep of Sec. VIII, which this pairing survives to  $\alpha^* = 3.55$  against the measured  $2.15$ – $3.15$ . (ii) *Readout*:  $d-1 = 2$  sequential shelving rounds at  $\sim 3 \times 10^{-3}$  worst-case misassignment each, plus  $\sim 3$  ms of detection and re-cooling idle per round [1]—the largest uncharged term, a few percent per idling carrier at the shielded  $\sim 100$  ms coherence unless echoed, which is precisely the suppression the  $\varepsilon^* = 0.09$ – $0.15$  specification prices. (iii) *Cross-talk*: residual  $\sim 2 \times 10^{-3}$  per gate [1], largely coherent (state-dependent AC-Stark shifts on occupied higher levels)—excluded from our channels (Limitations) and in principle calibratable, so it enters as a calibration requirement rather than a decoherence rate. (iv) *Motional heating*:  $2.7(2)$  phonons/s with  $27.4(4)$  ms motional coherence [48] against a  $\sim 10$  ms schedule—percent-level at most. Summed conservatively, the uncharged terms stay below the  $0.08$  raw margin of the  $m=4$  pair provided the collective- $B$  suppression of Sec. VIII is met; the  $m=3$  dead-heat pair is the systematics control that would expose any of them. For the ququint the table stands as a falsifiable *conditional* prediction: at matched per-base strength  $s = 10^{-3}$  the  $m = 3$  pair separates  $0.855$  vs  $0.778$ , and the separation widens down the demonstrated-fidelity band ( $0.575$  vs  $0.328$  at  $5 \times 10^{-3}$ )—but reaching the regime where both bases actually sit at a common  $s$  requires the  $96.7\%$  gate target above, a concrete specification for the next device generation rather than a run for the current one. The deep-circuit coherent failure of Sec. IX counsels entering at the shallow  $m = 2$  pair in either branch.

*Relation to code-level comparisons.* Keppens *et al.* [31] find slightly *worse* qudit logical error rates for the five-



TABLE X. Predictions for the proposed trapped-ion experiment: eigenstate QPE under per-particle depolarizing noise, **ion** cost model, exact density-matrix evolution (no sampling error). Strengths span the projected single-qudit figure ( $s = 10^{-3}$ ) and the band that *demonstrated* two-qudit entangling gates imply ( $5 \times 10^{-3}$ – $2 \times 10^{-2}$ : native gate fidelities of 93.7–98.7% at  $d = 3$ –5 [32]); a common  $s$  across bases is a matched-hardware idealization—the per-base entry point implied by the measured gate set is given in the text. Pairs are approximately matched in control dimension  $D$  (25 vs 32, 125 vs 128) and scored as the phase correct to  $b$  bits ( $b = 4$  is the finest criterion the  $D = 25$  grid can express). “gates” counts entangling gates (controlled- $U$  plus QFT rotations), with total entangling time-layers under the **ion** cost model in parentheses: a controlled- $U$  spans the  $w$ -digit work register and is charged  $w$  base layers, each entangling gate then multiplied by the  $(d-1)$  layer charge—e.g.  $d=5, m=3$ :  $3 \times (2 \cdot 4) + 3 \times 4 = 36$ ;  $d=2, m=7$ :  $7 \times 4 + 21 \times 1 = 49$ . Here  $w = \lceil \log_d 16 \rceil$  (4 at  $d=2$ , 2 at  $d=5$ ): the simulated target is a 16-dimensional diagonal-unitary system embedded in the work register (Sec. II), a genuinely multi-digit workload—unlike the hardware circuits of Sec. IX, where the work register is reduced to the single digit carrying the eigenstate.  $s$  is the per-carrier-layer strength of Sec. II. Readout is not charged; sequential-shelving readout adds up to  $d - 1$  detection rounds at  $\sim 3 \times 10^{-3}$  worst-case qudit error [1], a  $d$ -dependent cost the experimental budget must carry.

| register   | $b$ | gates   | noiseless | $s=0.001$ | 0.005 | 0.01  | 0.02  |
|------------|-----|---------|-----------|-----------|-------|-------|-------|
| $d=5, m=2$ | 4   | 3 (20)  | 0.816     | 0.785     | 0.674 | 0.562 | 0.401 |
| $d=2, m=5$ | 4   | 15 (30) | 0.917     | 0.807     | 0.500 | 0.299 | 0.142 |
| $d=5, m=3$ | 5   | 6 (36)  | 0.948     | 0.855     | 0.575 | 0.364 | 0.168 |
| $d=2, m=7$ | 5   | 28 (49) | 0.989     | 0.778     | 0.328 | 0.140 | 0.053 |

qudit code under the same noise convention we use, and the two results are compatible: a code is five carriers and a fixed gate list at every  $q$ —no problem instance to compress—so raising  $q$  buys no width or depth while charging every carrier more. An algorithm compresses; that compression is the entire source of the advantage measured here. A qudit advantage must be argued at the level of a compressible workload, and does not transfer automatically to the error-correction layer. Relatedly, Bocharov *et al.* [19] show emulated-binary arithmetic can beat native ternary—encoding density alone is not a mechanism—which is consistent with our attribution of the advantage to width-and-depth compression under a per-carrier noise budget, and with its disappearance when compilation buys the compression back.

*Recommendations to the field.* Cross-dimension algorithm comparisons should (i) randomize or report grid alignment (an irrational target phase does it for QPE; for order finding, choose  $r$  dividing no  $d^m$  and report residual misalignment); (ii) match problem size across bases by interpolation rather than comparing raw  $d^m$  points; (iii) charge multi-qudit gates their decomposition depth; and (iv) report the floor-to-baseline span of the metric (we require  $>0.15$ ; Sec. II discloses our one grandfathered exception), since small orders collapse the continued-

fraction metric’s dynamic range. Each rule was learned by stepping on the corresponding rake.

*Limitations.* Our channels are incoherent: coherent errors (cross-Kerr shifts  $\alpha \approx 0.1$ – $0.7$  MHz, drive-induced shifts) and leakage *during gates* are not modeled—idle thermal leakage now is (Sec. VIII), and helps the qudits. Non-Markovian and correlated dephasing enter through the quasi-static and common-mode controls of Sec. VIII, on both channel families. Cross-Kerr in particular is known to be fast on unprotected qutrits—at the measured  $\alpha \approx 0.1$ – $0.7$  MHz a residual always-on coupling dephases an unprotected two-qutrit coherence on the microsecond scale, i.e. within a few 580-ns gate durations [3]—and since the number of level pairs it can act on grows with  $d$ , its omission plausibly favors qudits—the sign of this bias runs against our conclusion, not for it. Gate cost and gate error are treated as proportional (more layers = more idle decoherence); a model where longer gates achieve better per-operation fidelity would soften the ion penalty, and the inflation threshold of Sec. VIII bounds the tolerance in the opposite direction. The schedule is serial (every carrier idles during every gate), matching single-addressed ion strings; platforms that execute gates concurrently cut idle exposure most for the widest register, i.e. for  $d = 2$ . That bias is now measured rather than asserted (Sec. VIII): under an exact as-soon-as-possible schedule one cell of Table II changes verdict (the thin ladder/*pavlidis* qutrit win), the rest hold with margins shrunk by up to  $4\times$ . The scaling trends are measured to Hilbert-space dimension  $5.3 \times 10^5$  and stated as slopes with error bars, not theorems: the Shor sweep uses two instances ( $N = 21$ ,  $r = 6$  and  $N = 29$ ,  $r = 7$ ) at one noise strength per regime, with 3–6 sizes per base. The ordering and the qubit’s fastest decay replicate across the instances; the qutrit’s *rate* replicates only in the ladder channel ( $1.7\sigma$  over matched sizes) and not under depolarizing ( $3.7\sigma$ ), so individual per-base slopes carry instance-level variance of order 0.03/bit—larger than the earlier size-mismatched comparison suggested. The qutrit families are plateau-then-fall rather than linear, so their quoted slopes average a shape: six sizes locate the fall in the last three but cannot fit its onset, and the  $N = 29$  sweep stops before reaching it. Whether the plateau lengthens, or the fall steepens, beyond 14.3 bits is unmeasured. The qutrit’s decoder account is derived and verified exactly (Sec. VII C) and grid alignment is excluded as a confound. Statistical error bars are small throughout; the dominant uncertainty is the systematic spread across channel and cost-model choices, which is why we report the full grid. Dimensions beyond  $d = 5$  are tested only at demo size on one instance ( $d = 7, 11, 13$ ; Sec. IV), where the linear-cost window closes by  $d = 11$  and the native-gate cells are dominated by register granularity—no scaling sweep exists above  $d = 5$ . All simulations are of ideal algorithm circuits: the controlled- $U^{d^i}$  multipliers are applied as exact unitaries, each charged one gate at the cost model’s per-gate rate, whereas com-



piled modular arithmetic—depth  $4d^2q$  per adder under two-level decomposition [16]—has a  $d$ -dependence of its own: at face value the compiled qudit-to-qubit depth ratio spans  $\approx 0.9$ – $2.7$  across  $d = 3, 5$  (depending on whether adder- or full-exponentiation depths dominate), extending well beyond the pavlidis row’s 0.85–1.09, so compiled arithmetic can be harsher than any cost model charged here and the decomposition verdict is, if anything, conservative. Whether the penalty stays confined to depth is only partially settled: the follow-up construction of Ref. [10] gives an in-place, ancilla-free multiplier of width exactly  $n$  with nearest-neighbor interactions—but for multiplication by a *constant* modulo the register size  $p^n$  (unit constants only), not the mod- $N$  modular multiplication Shor’s controlled- $U^{d^i}$  requires, which in general needs comparison-and-correction machinery with ancillas of its own. Within that scope compiled arithmetic inflates the layers term of the carriers  $\times$  layers exposure budget (Sec. VII) without touching the carrier count and maps onto a linear ion chain without routing overhead; the width cost of full mod- $N$  arithmetic on qudits remains open. Fault-tolerant overhead is out of scope, and per Ref. [31] its dimension-dependence may differ in sign. Finally,  $d = 5$  entangling gates exist today only on trapped ions [32]—our deep-register results are one to two hardware generations ahead of experiment, and Sec. VIII quantifies how far the measured gate fidelities sit from the advantage regime.

*Summary.* In bare, uncorrected circuits the qudit question has a determinate answer, and it is doubly conditional. Width-and-depth compression is the entire mechanism of the advantage: a native entangling gate of at-most-linear cost preserves it; two-level decomposition spends it back in proportion to dimension—fully at  $d \geq 5$  on the ladder and at  $d = 7$  everywhere, only partly at  $d = 3$ , where the corrected charge leaves a thin advantage that per-particle noise sustains and realistic ladder accounting erases; the native gate’s own infidelity growth with  $d$  can spend it back just as surely—at measured fidelities it does, for the ququint, in seven of eight readings—and Markovian-idealized structured dephasing overwhelms it, though its laboratory quasi-static counterpart does not. What the algorithm itself adds is classical—a decoder whose error tolerance obeys an exact mediant-interval law with totient-sum asymptotics, identical across bases at matched control dimension—so the quantum state carries the whole cross-base difference. Every cross-dimension benchmark inherits these terms. The cheapest decisive test on current hardware is eigenstate QPE at  $d = 3$  on a trapped-ion qudit processor—a pairing measured fidelities leave alive under both inflation scopes—while the  $d = 5$  test waits on a native entangling gate at  $\gtrsim 96.7\%$  fidelity, a concrete target 3 percentage points above the demonstrated state of the art.

## XI. METHODS

*Simulators.* Exact density-matrix evolution (states as rank- $2n$  tensors, channels applied per carrier per layer) to Hilbert dimension  $\sim 3000$ ; beyond that, Monte Carlo wavefunction trajectories [53, 54]: after each gate, each carrier independently passes through the per-layer channel raised to the gate’s cost, sampled via one Kraus operator drawn with probability  $\text{tr}(K^\dagger K \rho_q)$  from the carrier’s reduced state. Averaging  $|\psi\rangle\langle\psi|$  over trajectories reproduces the channel exactly; each trajectory contributes its full outcome distribution, so statistical error is well below Bernoulli—measured as a variance ratio  $p(1-p)/\text{var}(S_k)$  of 7.7–26.6 over the five points characterized below. That estimator is checked rather than asserted, on both counts a  $\sigma$ -claim depends on. Unbiasedness: at  $d = 3$ ,  $m = 4$  under the calibrated ladder, 8000 trajectories give  $0.24294 \pm 0.00092$  against an exact density matrix’s 0.242474, a  $0.51\sigma$  agreement. Calibration: 24 independent 1000-trajectory replicas at each of  $(d, m) = (3, 4), (3, 6), (2, 8), (5, 3)$  put the empirical standard error of the mean against the quoted one at a ratio of 0.995 (range 0.84–1.17; smallest Bartlett  $p = 0.22$ ), with realized  $1\sigma$  coverage 0.58–0.79 around the nominal 0.68. The quoted bars are therefore neither conservative nor optimistic, and every  $\sigma$  and  $\chi^2$  in Sec. V is computed from them by a committed script rather than by hand.

*Reproducible seeds.* Every trajectory sweep now seeds from `crc32` of its configuration tuple. It previously seeded from `hash` of that tuple, and CPython salts string hashing per process, so those seeds—and therefore the exact runs behind an earlier release of these numbers—could not be regenerated. All sweeps were re-run after the change. The re-run doubles as the broadest calibration test in this work: old and new draw independent streams at identical settings, so if the quoted standard errors are honest the shifts  $z = (\text{new} - \text{old})/\sqrt{\sigma_{\text{new}}^2 + \sigma_{\text{old}}^2}$  must be standard normal. Over the 368 paired points they give mean  $z = -0.09 \pm 0.05$  and  $\text{sd } z = 0.98 \pm 0.04$ , with 5.2% beyond  $2\sigma$  against 4.6% expected—agreement over the whole grid, not just the five points characterized above. Two conclusions moved, and both are stated in their corrected form in the text: one of the six grid-alignment predictions now misses by  $1.4\sigma$  (Sec. III), and the qutrit’s depolarizing slope no longer replicates across instances (Sec. V). Every other quoted trajectory number was re-synced to the post-reseed run for this revision; none of those changes alters a verdict. No cost-table or robustness verdict changed—those are exact density matrices and carry no seed. Every paired shift is recorded in `results/prehash_comparison.json`; the size-scaling fits and the qutrit/ququint crossings are regenerated by `scaling_claims.py`, and the entanglement-interpolation slopes by `interpolation_slopes.py`. Separately from the re-seeding, an earlier version of the cost-model code charged single-qudit gates the  $d^2/4$  pavlidis multiplier that both the text and Ref. [16]’s construction reserve for controlled rotations; the corrected charge (single-

qudit gates one layer) re-scores every `pavlidis` cell—Table II, the  $d = 7$  grid, eigenstate QPE, Grover, and the concurrency and single-qudit-cost analyses—and every such number in this version reflects it. Fractional gate costs are exact: both channel families form one-parameter semigroups, so  $E^t$  is evaluated by scaling the Lindblad rate (ladder) or as  $1 - q = (1 - p)^t$  (depolarizing). End-state fidelity is estimated as the trajectory average of  $|\langle \psi_{\text{ideal}} | \psi \rangle|^2$ , an unbiased estimator of  $\langle \psi_{\text{ideal}} | \rho | \psi_{\text{ideal}} \rangle$ , cross-checked against exact density-matrix evolution. Throughout,  $F_e = \text{tr } S/d^2$  with  $S$  the one-layer superoperator in the natural representation; for a channel with Kraus operators  $\{K_i\}$  this is  $\sum_i |\text{tr } K_i|^2/d^2$ , the entanglement fidelity of the channel with respect to the maximally entangled state.

*Calibrated dephasing via Euclidean embedding.* With diagonal jump operators  $\{J\}$ , pairwise dephasing rates satisfy  $\Gamma_\phi(j, k) = \frac{1}{2} \|v_j - v_k\|^2$  where  $v_j$  collects the  $j$ -th diagonal entries across jumps; realizing a target  $\Gamma_\phi$  matrix is therefore a Euclidean embedding problem, solved exactly by classical multidimensional scaling (residual  $\leq 3 \times 10^{-17}$  for  $d = 2-7$ ). This realizes the measured max-level law exactly—something no linear-frequency-ladder model can do.

*Verification.* Twenty correctness tests: QFT circuits against dense matrices; complete positivity and trace preservation of every channel (Choi eigenvalues); trajectory-vs-exact agreement; fidelity-estimator cross-check; noiseless baselines; grid-alignment predicates; unbiased-instance orderings; Grover correctness and cost accounting; and readout-channel monotonicity. The exact decoder count of Sec. VII C and the formula of Eq. (3) are verified by outcome-for-outcome comparison against decoder enumeration on 27 instance/size combinations (the 11 register sizes of Table IV; 16 within-modulus (order,  $D$ ) pairs on  $N = 33, 55$ ). The external reproduction of Ref. [34] (Sec. VIII) runs separately from the test suite. Statistics: 1000 trajectories per Shor and QPE scaling point; 400 per point for the  $N = 29$  instance-robustness sweep (200 at the three deepest configurations, whose work registers are wider than at  $N = 21$ ); 400 trajectories per point for the grid-alignment and within-modulus studies; 1000 for the eigenstate-interpolation study and for the  $d = 7$  demo grid; 200 for the fidelity-collapse Shor points (100 at  $d=2$ ,  $m=12$ ), whose Grover points use exact density-matrix evolution; demo-size cost, SPAM, and refocusing results use exact density-matrix evolution throughout, as does the multiplicative-group ensemble over all eleven units of  $\mathbb{Z}_{21}^\times$  (Sec. III); its companions over all nineteen units of  $\mathbb{Z}_{33}^\times$  and all thirty-nine of  $\mathbb{Z}_{55}^\times$  use 400 trajectories per point. Error bars are standard errors over trajectories; cross-base differences quoted with uncertainties are propagated through the interpolation onto the common size axis.

*Hardware runs.* Via AWS Braket (August 2026). IonQ Forte-1: 1,000-shot raw pilots, a 500-shot  $m = 4$  probe, and 5,000-shot main runs with IonQ debiasing; IonQ’s native  $ZZ$  accepts  $|\theta| \leq \pi/2$ , so each controlled-phase

compiles as  $R_z(\theta/2) \otimes R_z(\theta/2) \cdot ZZ(-\theta/2)$  with any excess folded into paired  $R_z(\pi)$  rotations—every compiled variant reproduces the analytic outcome distribution on the Braket local simulator to  $10^{-15}$  before submission. IQM Garnet: 5,000 shots per circuit, server-side routing, native controlled-phase support. Success windows, floors, and the digit-reversed scoring are identical to the simulation pipeline; per-task records, raw outcome histograms, and scoring code ship with the repository.

*Reproducibility.* All code, calibration fits, result files, and the exact scripts that produce every figure and table are available at <https://github.com/doxaras/qudit-decoherence>; the v1.0 snapshot is archived at doi:10.5281/zenodo.21901533, and the repository carries the current version, including the measured-fidelity re-analysis of Sec. VIII.

## Appendix A: The decoder acceptance lemma

The decoder used throughout this paper is the following procedure. On outcome  $y \in [1, D)$ , scan the continued-fraction convergents  $p/q$  of  $y/D$  in order of increasing denominator; at the first  $q \leq N$  with  $a^q \equiv 1 \pmod{N}$ , return the least  $r'$  dividing  $q$  with  $a^{r'} \equiv 1 \pmod{N}$ ; if no convergent denominator  $q \leq N$  passes the test, reject ( $y = 0$  is always rejected).

*Lemma.* The decoder returns the order  $r$  of  $a$  modulo  $N$  if and only if some convergent denominator  $q$  of  $y/D$  satisfies  $q \leq N$  and  $r \mid q$ ; on every other outcome it rejects. In particular it never returns a wrong order.

*Proof.* Since  $r$  is the multiplicative order of  $a$  modulo  $N$ ,  $a^q \equiv 1 \pmod{N}$  if and only if  $r \mid q$ . ( $\Leftarrow$ ) Suppose some convergent denominator  $q \leq N$  is a multiple of  $r$ ; the scan stops at the first such  $q$ . Every divisor  $r'$  of  $q$  with  $a^{r'} \equiv 1$  is itself a multiple of  $r$ , and  $r$  both divides  $q$  and passes the test, so the least such  $r'$  is exactly  $r$ . ( $\Rightarrow$ ) If no convergent denominator  $q \leq N$  is a multiple of  $r$ , the test  $a^q \equiv 1$  never fires and the decoder rejects.  $\square$

The equivalence is verified empirically by `decoder_formula.py` in the repository: enumerating the decoder over every outcome  $y \in [0, D)$  and comparing against the convergent predicate gives bit-identical acceptance sets on 42 (instance,  $D$ ) combinations—six instances ( $N = 21, 29$ , and the four within-modulus orders of  $N = 33, 55$ ) across seven control dimensions each.

*Relation to the divisor form.* Standard analyses certify a different event: for  $y$  within  $1/(2\tilde{r}^2)$  of a peak  $s/r$ , the convergent guarantee [37] yields the reduced fraction  $(s/g)/(r/g)$  with  $g = \text{gcd}(s, r)$ , whose denominator  $\tilde{r} = r/g$  divides  $r$  [25, 41]. A divisor with  $g > 1$  fails the test  $a^{\tilde{r}} \equiv 1$ , so those analyses lift  $\tilde{r}$  to  $r$  by classical search over multiples (or over gcd-smoothness classes [41]). The decoder above performs no lift: it succeeds precisely through the convergent denominators that are already multiples of  $r$ —including the deeper convergents  $kr \leq N$ ,  $k \geq 2$ , outside the sufficient window—which is why its

acceptance set obeys the totient sum of Eq. (5) over the admissible denominators  $r, 2r, \dots, \lfloor N/r \rfloor r$  rather than a window count.

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*AI-use disclosure.* The simulation and analysis code was generated with substantial assistance from large-language-model tools, directed and reviewed by the author, who takes full responsibility for the content; every quantitative claim is regenerable from the committed scripts named in the text.

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